MECHANICAL SYMBOLS AND ABBREVIATIONS

NOTE: ALL SYMBOLS AND ABBREVIATIONS SHOWN ARE NOT NECESSARILY USED ON THE DRAWINGS

GENERAL NOTES

1. PERFORM ALL WORK IN ACCORDANCE WITH ALL APPLICABLE CODES AND AUTHORITIES HAVING JURISDICTION. PROVIDE ALL PERMITS, INSPECTIONS, LICENSES AND FEES. FURNISH ALL LABOR, EQUIPMENT, SUPPLIES AND MATERIALS NECESSARY TO PROVIDE COMPLETE AND OPERATIONAL SYSTEMS.

2. THE DRAWINGS AND SPECIFICATIONS INDICATE THE GENERAL DESIGN AND ARRANGEMENT OF PIPES, FIXTURES, EQUIPMENT, SYSTEMS, ETC. INFORMATION SHOWN IS DIAGRAMMATIC IN CHARACTER AND DOES NOT NECESSARILY INDICATE EVERY REQUIRED OFFSET, FITTING, ETC. DO NOT SCALE THE DRAWINGS FOR DIMENSIONS. TAKE ALL DIMENSIONS, MEASUREMENTS, ETC., FROM THE EQUIPMENT TO BE FURNISHED. PIPING MAY BE RELOCATED OR OFFSET FOR PROPER CLEARANCES OR TO AVOID CONFLICTS WITH OTHER TRADES. THE DESIGN INTENT (I.E. PITCHES, VELOCITIES, PRESSURE DROPS, VOLTAGE DROPS, ETC) CANNOT BE GREATLY ALTERED WITHOUT THE APPROVAL OF THE ENGINEER. THE COST OF THESE DEVIATIONS TO AVOID INTERFERENCE'S SHALL BE PART OF THE ORIGINAL CONTRACT BID.

3. EACH SUBCONTRACTOR SHALL CONFER AND COOPERATE WITH ALL OTHER TRADES TO COORDINATE THEIR WORK. COORDINATION SHALL INCLUDE, BUT SHALL NOT BE LIMITED TO MATERIALS AND EQUIPMENT ROUTED IN CEILING AND WALL CAVITIES, EQUIPMENT ARRANGEMENT IN MECHANICAL SPACES, INCLUDING EQUIPMENT CLEARANCE REQUIREMENTS, ELEVATIONS AND DIMENSIONS OF STRUCTURAL MEMBERS AND OPENINGS, ETC. THE CONTRACTOR SHALL NOTIFY THE ENGINEER OF ANY CONFLICTS.

4. BASE FINAL INSTALLATION OF MATERIALS AND EQUIPMENT ON ACTUAL DIMENSIONS AND CONDITIONS AT THE PROJECT SITE. FIELD MEASURE FOR MATERIALS AND EQUIPMENT REQUIRING EXACT FIT. NO EXTRAS WILL BE GIVEN FOR THE CONTRACTORS FAILURE TO FIELD COORDINATE.

5. THE OWNER OR ENGINEER ARE NOT RESPONSIBLE FOR THE CONTRACTOR'S SAFETY PRECAUTIONS OR FOR MEANS, METHODS, TECHNIQUES, CONSTRUCTION SEQUENCES, OR PROCEDURES REQUIRED TO PERFORM THE WORK.

6. THE CONTRACTOR SHALL LOCATE ALL EQUIPMENT THAT MUST BE SERVICED, OPERATED, OR MAINTAINED IN FULLY ACCESSIBLE POSITIONS. EQUIPMENT SHALL INCLUDE (BUT NOT LIMITED TO) VALVES, MOTORS, CONTROLLERS, SWITCHGEAR, AND DRAIN POINTS IF REQUIRED FOR BETTER ACCESSIBILITY. FURNISH ACCESS DOORS FOR THIS PURPOSE. MINOR DEVIATIONS FROM THE DRAWINGS MAY BE ALLOWED TO PROVIDE FOR BETTER ACCESSIBILITY. ANY CHANGES SHALL BE APPROVED BY THE ENGINEER AND CONSTRUCTION MANAGER/GENERAL CONTRACTOR PRIOR TO MAKING THE CHANGE.

7. THE CONTRACTOR SHALL PROVIDE ACCESS DOORS, WALL OPENINGS, ROOF OPENINGS OR ANY OTHER CONSTRUCTION REQUIREMENT NEEDED TO ACCOMMODATE THE MECHANICAL EQUIPMENT. LOCATIONS OF THESE OPENINGS SHALL BE SUBMITTED IN SUFFICIENT TIME TO BE INSTALLED IN THE NORMAL COURSE OF WORK.

8. THE CONTRACTOR SHALL COORDINATE ELECTRICAL REQUIREMENTS OF APPROVED EQUIPMENT WITH THE ELECTRICAL CONTRACTOR PRIOR TO THE PURCHASE AND INSTALLATION OF ANY ELECTRICAL GEAR OR CONDUIT.

9. GENERAL CONTROL WIRING, THERMOSTATS, MOTORIZED DAMPERS AND CONDUIT ASSOCIATED WITH HVAC EQUIPMENT IS THE RESPONSIBILITY OF THE CONTRACTOR. THE CONTRACTOR SHALL COORDINATE THE LOCATION OF ALL THERMOSTATS, ROOM SENSORS, ETC WITH THE ENGINEER AND ALL OTHER TRADES PRIOR TO INSTALLATION. IF A CONFLICT WITH MILLWORK, LIGHT SWITCHES, WINDOWS, ETC EXISTS, THE CONTRACTOR SHALL NOTIFY THE ENGINEER OF THE POTENTIAL INTERFERENCE PRIOR TO INSTALLATION. INSTALL THERMOSTATS WITH PROTECTIVE LOCKING COVER, CENTERED AT 4'-0" ABOVE FINISHED FLOOR, UNLESS OTHERWISE INDICATED. COMPLY WITH THE PROVISIONS OF THE AMERICANS WITH DISABILITIES ACT (ADA) AND THE TEXAS ACCESSIBILITY'S STANDARD (TAS).

10. ALL DIMENSIONS SHOWN ON THE DRAWINGS FOR DUCTWORK ARE NET INSIDE CLEAR DIMENSIONS. FOR RECTANGULAR DUCT, THE FIRST FIGURE OF THE DUCT SIZE INDICATES THE DIMENSION OF THE FACE SHOWN. VERIFY THAT THE DUCTWORK SPECIFIED WILL FIT IN THE SPACE AVAILABLE USING THE FIELD VERIFIED DIMENSIONS OF THE EXISTING CONDITIONS (REFER TO DEMOLITION WORK NOTES THIS SHEET) PRIOR TO FABRICATION AND INSTALLATION.

11. PROVIDE VIBRATION ISOLATORS FOR MOTOR DRIVEN EQUIPMENT UNLESS NOTED OTHERWISE. PROVIDE ISOLATION AS INDICATED OR AS RECOMMENDED BY THE EQUIPMENT MANUFACTURER.

12. SOME PIPES AND DUCTS SHOWN ON EACH FLOOR PLAN MAY BE SHOWN WITH AN OFFSET FOR CLARITY.

13. SEAL ALL PIPE AND DUCT PENETRATIONS THROUGH FIRE RATED BUILDING ELEMENTS WITH AN APPROVED FIRE PROOFING MATERIAL.

14. ALL EQUIPMENT SHALL HAVE IDENTIFICATION TAGS. TAGS SHALL BE PLASTIC LAMINATE, WHITE FACE WITH 1/2" TALL BLACK LETTERS. THE TAG SHALL MATCH THE UNIT DESIGNATIONS SHOWN ON THE SCHEDULES. EACH FAN IN EACH FAN WALL ARRAY SHALL HAVE A UNIQUE LABEL (E.G. SF1-1, SF1-2, ETC.).

15. WHEN ALL WORK IS COMPLETE, REFINISH/RESTRIPE FLOORS AND PATCH AND REPAINT WALLS TO BE LIKE NEW CONDITION. WHEN ALL PIPING WORK IS COMPLETE PAINT NEW AND EXISTING PIPING CHILLED WATER PIPING IN ACCORDANCE WITH OWNER/COUNTY COLOR STANDARDS.

16. COORDINATE ALL CONTROLS WORK WITH RELIABLE CONTROLS (CONTACT: SID ELLIS, sellis@enviromaticsystems.com, (972)206-2635)

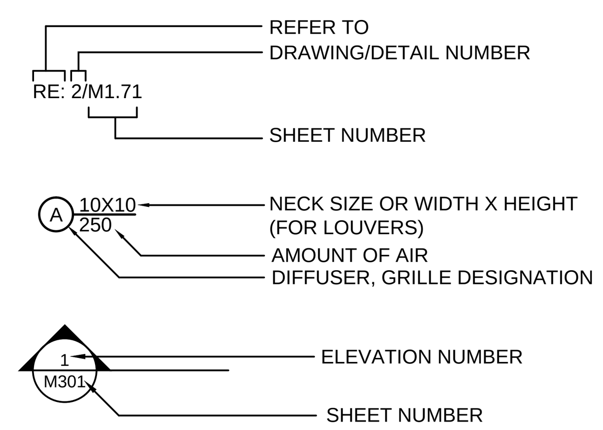
ABBREVIATIONS

AD	ACCESS DOOR	L	LENGTH
A/C	AIR CONDITIONING UNIT	LAT	LEAVING AIR TEMPERATURE
A/E	ARCHITECT/ENGINEER	LPC	LOW PRESSURE CONDENSATE
AFF	ABOVE FINISHED FLOOR	LPS	LOW PRESSURE STEAM
AFS	AIR FLOW SWITCH	LB	POUNDS
AHU	AIR HANDLING UNIT	LRA	LOCKED ROTOR AMPS
APPROX	APPROXIMATE	LWT	LEAVING WATER TEMPERATURE
BAS	BUILDING AUTOMATION SYSTEM	MAX	MAXIMUM
BHP	BRAKE HORSE POWER	MBH	1000 BRITISH THERMAL UNITS / HOUR
BTU	BRITISH THERMAL UNIT PER HOUR	MCA	MINIMUM CIRCUIT AMPACITY
C/A	COMBUSTION AIR	MFR	MANUFACTURER
CC	COOLING COIL	MIN	MINIMUM
CFH	CUBIC FEET PER HOUR	N/A	NOT APPLICABLE
CFM	CUBIC FEET PER MINUTE	N/O,N/C	NORMALLY OPEN, NORMALLY CLOSED
CLG	CEILING	O/A	OUTSIDE AIR/FRESH AIR
CU	CONDENSING UNIT	OBD	OPPOSED BLADE DAMPER
D	EQUIPMENT DRAIN	O/C	ON CENTER
DEG	DEGREES	PEF	PURGE EXHAUST FAN
DB	DRY BULB	PH	PHASE
DN	DOWN	PROVIDE	FURNISH AND INSTALL
(E)	EXISTING	PRV	PRESSURE REDUCING VALVE
EAT	ENTERING AIR TEMPERATURE	PSI	POUNDS PER SQUARE INCH
E/A	EXHAUST AIR	R/A	RETURN AIR
EDH	ELECTRIC DUCT HEATER	RE:	REFERENCE, REFER
EF	EXHAUST FAN	RL	REFRIGERANT LIQUID
ESP	EXTERNAL STATIC PRESSURE	RLA	RUNNING LOAD AMPS
EWT	ENTERING WATER TEMPERATURE	RM	ROOM
°F	DEGREES FAHRENHEIT	RPM	REVOLUTIONS PER MINUTE
FCU	FAN COIL UNIT	RS	REFRIGERANT SUCTION
FD	FIRE DAMPER	S/A	SUPPLY AIR
FLA	FULL LOAD AMPS	SD	SMOKE DETECTOR
FLR	FLOOR	SF	SQUARE FOOT, SUPPLY FAN
FPVAV	FAN POWERED VAV	SPECS	SPECIFICATIONS
FSD	FIRE SMOKE DAMPER	T, TSTAT	THERMOSTAT, ROOM SENSOR
FT.	FOOT, FEET	T/A	TRANSFER AIR
FT. WG	FEET WATER GAUGE	THRU	THROUGH
GA	U.S. GAUGE	TSP	TOTAL STATIC PRESSURE
GPM	GALLONS PER MINUTE	TSTAT	THERMOSTAT OR ROOM SENSOR
H	HEIGHT	TYP	TYPICAL
HP	HORSEPOWER	UL	UNDERWRITERS LABORATORIES, INC.
HPC	HIGH PRESSURE CONDENSATE	UH	UNIT HEATER
HPS	HIGH PRESSURE STEAM	V	VOLTS
HWR	HEATING WATER RETURN	VAV	VARIABLE AIR VOLUME
HWS	HEATING WATER SUPPLY	VEL	VELOCITY
HZ	HERTZ	VFD	VARIABLE FREQUENCY DRIVE
IN.	INCH, INCHES	W/	WITH
IN.WG	INCHES WATER GAUGE	WB	WET BULB
J-BOX	JUNCTION BOX	W/O	WITHOUT
kW	KILOWATT		

LINE TYPES

SYMBOL	DESCRIPTION
— CS —	CONDENSER WATER SUPPLY
— CR —	CONDENSER WATER RETURN
— CCCS —	COMPUTER CONDENSER WATER SUPPLY
— CCCR —	COMPUTER CONDENSER WATER RETURN
— CHS —	CHILLED WATER SUPPLY
— CHR —	CHILLED WATER RETURN
— CCHS —	COMPUTER CHILLED WATER SUPPLY
— CHR —	COMPUTER CHILLED WATER RETURN
—HWS—	HEATING WATER SUPPLY
—HWR—	HEATING WATER RETURN
—RD—	REFRIGERANT DISCHARGE
—RS—	REFRIGERANT SUCTION
—RL—	REFRIGERANT LIQUID
—HPS—	HIGH PRESSURE STEAM
—HPC—	HIGH PRESSURE CONDENSATE
—LPS—	LOW PRESSURE STEAM
—LPC—	LOW PRESSURE CONDENSATE
—PC—	PUMPED CONDENSATE
—MU—	MAKE-UP WATER
	DIRECTION OF FLOW
	DIRECTION OF PIPE SLOPE DOWN

DRAWING/DETAIL REFERENCE



MISCELLANEOUS

1	DRAWING NOTE REFERENCE (I.E., NOTES BY SYMBOL)
	CONNECTION INTO EXISTING

SYMBOLS

SYMBOL	DESCRIPTION
	SQUARE CEILING DIFFUSER (SUPPLY) (4-WAY UNLESS OTHERWISE INDICATED)
	SQUARE CEILING GRILLE (RETURN OR EXHAUST)
T TS	THERMOSTAT (OR) TEMP SENSOR
M	MOTORIZED DAMPER
or	MANUAL VOLUME DAMPER
FD	FIRE DAMPER

VALVES AND FITTINGS

SYMBOL	DESCRIPTION
	SHUT-OFF / ISOLATION VALVE
	BALL VALVE
	BUTTERFLY VALVE
	GLOBE VALVE
	PLUG VALVE / COCK VALVE
	CHECK VALVE
	2-WAY CONTROL VALVE
	3-WAY CONTROL VALVE
	SOLENOID VALVE
	STRAINER
	CALIBRATED BALANCING VALVE
	FLOW SWITCH
	UNION (DIELECTRIC)
	VALVE IN RISER
	END RISE (90° ELL)
	END DROP (90° ELL)
	RISE OR DROP
	TEE OUT OF TOP OF PIPE
	TEE OUT OF BOTTOM OF PIPE
	CAP ON END OF PIPE
AG	ALIGNMENT GUIDE
	PIPE ANCHOR
	PIPE DEMOLITION

BASIS OF MECHANICAL DESIGN

PRIMARY MECHANICAL CODES:

MECHANICAL: 2009 INTERNATIONAL MECHANICAL CODE (WITH CITY AMENDMENTS).
ENERGY: 2009 INTERNATIONAL ENERGY CODE (WITH CITY AMENDMENTS).

PROJECT DESIGN VALUES:

OUTDOOR DESIGN TEMPERATURE (SUMMER):	100.6°F (DRYBULB), 74.6°F (WETBULB)
AMBIENT TEMPERATURE AT CONDENSING UNITS:	105°F (DRYBULB, SUMMER)
OUTDOOR DESIGN TEMPERATURE (WINTER):	21.9°F (DRYBULB)
INDOOR DESIGN TEMPERATURE (SUMMER):	75°F (DRYBULB), 50% (RELATIVE HUMIDITY)
INDOOR DESIGN TEMPERATURE (WINTER):	70°F (DRYBULB)
OUTSIDE AIR REQUIREMENTS:	PER IMC TABLE 403.3

DEMOLITION WORK NOTES

GENERAL

1. EXISTING WORK SHOWN ON PLANS IS FROM PREVIOUS ENGINEERING DOCUMENTS AND FIELD OBSERVATIONS. ACTUAL CONDITIONS MAY VARY, AND CONTRACTOR MUST FIELD VERIFY EXISTING WORK AND MAKE MINOR ADJUSTMENTS NECESSARY TO COMPLETE WORK. IF EXISTING CONDITIONS PROHIBIT WORK, NOTIFY THE ENGINEER FOR DIRECTION, AS REQUIRED.

2. COORDINATE WITH ALL TRADES FOR REQUIRED CEILING REMOVAL IN EXISTING BUILDING. NOTIFY THE ENGINEER AND OWNER PRIOR TO COMMENCING REMOVAL. REMOVE ONLY THAT PORTION OF THE CEILING NECESSARY TO ACCESS AND COMPLETE THE WORK. UPON COMPLETION OF THE ABOVE CEILING WORK, CEILING IS TO BE REINSTALLED. REPLACE ANY DAMAGED CEILING TILES WITH NEW TILES TO MATCH EXISTING.

3. DEMOLITION SHALL EXTEND TO POINTS OF CONNECTION WITH LIVE SERVICES (PANELBOARDS, PIPING MAINS, ETC). DEMOLITION SHALL NOT PERMIT ABANDONMENT OF ANY PORTION OF ANY SYSTEM UNLESS SPECIFICALLY NOTED AS "ABANDON IN PLACE" OR "TO REMAIN". DO NOT LEAVE ANY OF THE EXISTING CHILLED WATER PIPING OVER OR AROUND THE DISPATCH SERVER ROOM.

4. DEMOLITION SHALL INCLUDE EQUIPMENT, PIPING, DUCTWORK, SUPPORTS, FITTINGS, ACCESSORIES, CONTROLS, WIRING, CONDUIT, ETC, IN THEIR ENTIRETY UNLESS OTHERWISE NOTED.

5. IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO VERIFY THE CONDITION OF ALL EXISTING EQUIPMENT WITHIN THE PROJECT SCOPE, EXACT SIZES OF EXISTING DUCT AND PIPING, ETC BEFORE COMMENCING DEMOLITION WORK. REPORT ANY DISCREPANCIES BETWEEN PLANS AND ACTUAL FIELD CONDITIONS TO ENGINEER PRIOR TO THE COMMENCEMENT OF DEMOLITION WORK.

6. PATCH OPENINGS IN WALLS TO MAINTAIN THE INTEGRITY OF THE WALL WHERE AIR DEVICES HAVE BEEN REMOVED.

EQUIPMENT

1. THE OWNER HAS THE FIRST RIGHT-OF-REFUSAL FOR ALL DEMOLISHED EQUIPMENT. THE CONTRACTOR IS RESPONSIBLE FOR REMOVAL AND PROPER DISPOSAL OF ANY EQUIPMENT REFUSED BY THE OWNER.

2. ALL REMOVED EQUIPMENT SHALL BE MAINTAINED IN GOOD CONDITION. REMOVED EQUIPMENT NOT INDICATED FOR RE-USE SHALL REMAIN THE PROPERTY OF THE OWNER. THE CONTRACTOR SHALL REMOVE THE EQUIPMENT AND DELIVER IT TO THE OWNER. SHOULD THE OWNER DECLINE THE POSSESSION OF THE REMOVED EQUIPMENT, IT SHALL BECOME THE PROPERTY OF THE CONTRACTOR FOR REMOVAL FROM SITE.

3. WHEN ALL CONSTRUCTION IS COMPLETE INSTALL NEW, CLEAN PRE-/POST-FILTERS IN AIR UNITS SERVING THE RENOVATED AREAS. VERIFY CONDITION OF UNIT FILTER GAUGES AND REPAIR OR REPLACE IF FOUND TO BE DAMAGED OR NON-FUNCTIONAL.

DUCTWORK

1. CAP AND SEAL AIR TIGHT ALL POINTS AT WHICH DUCTWORK IS REMOVED FROM DUCTWORK THAT WILL REMAIN. RE-INSULATE REMAINING DUCTWORK TO MAINTAIN VAPOR BARRIER.

2. CONTRACTOR SHALL VERIFY CLEARANCE REQUIREMENTS AND INDICATE ROUTING OF NEW DUCTWORK BEFORE FABRICATION BEGINS AS RISES AND DROPS MAY BE NECESSARY DUE TO EXISTING FIELD CONDITIONS.

PIPING

1. WHERE PIPING IS SHOWN TO BE DEMOLISHED, IT SHALL BE DEMOLISHED TO THE POINT OF ORIGIN AT THE NEAREST ACTIVE MAIN. INSTALL SHUT-OFF VALVE AND CAP FOR FUTURE CONNECTION.

CONTROLS

1. DEMOLITION AND/OR RELOCATION OF CONTROLS FOR EQUIPMENT SHALL INCLUDE, BUT NOT BE LIMITED TO:
   SPACE AND DUCT THERMOSTATS
   SPACE AND DUCT TEMPERATURE/HUMIDITY SENSORS;
   SMOKE DETECTORS, FIRE-STATS, FREEZE-STATS, AND OTHER SAFETY OR LIMITING DEVICES;
   CONTROL SYSTEM PANELS

RELIEF FAN SCHEDULE

MARK EF-	TYPE	LOCATION	CFM	EXT. SP IN WG	MOTOR DATA: QTY	HP (EACH)	UNIT MCA	VOLTS	PH	DRIVE	MAX SONES	WEIGHT	MANUFACTURER/ MODEL SERIES	REMARKS
RF-1	DIRECT DRIVE FAN WALL	ROOF PENTHOUSE	15,000	1.0	9	1.0	12.6	460	3	DIRECT	15.0	3,000	GOVERNAIR/FANWALL	1,2,3,4
RF-2	DIRECT DRIVE FAN WALL	ROOF PENTHOUSE	15,000	1.0	9	1.0	12.6	460	3	DIRECT	15.0	3,000	GOVERNAIR/FANWALL	1,2,3,4
RF-3	DIRECT DRIVE FAN WALL	ROOF PENTHOUSE	15,000	1.0	9	1.0	12.6	460	3	DIRECT	15.0	3,000	GOVERNAIR/FANWALL	1,2,3,4
RF-4	DIRECT DRIVE FAN WALL	ROOF PENTHOUSE	15,000	1.0	9	1.0	12.6	460	3	DIRECT	15.0	3,000	GOVERNAIR/FANWALL	1,2,3,4

REMARKS:
1. PROVIDE WITH DISTRIBUTION PANEL (MSP) FOR SINGLE POINT ELECTRICAL CONNECTION FOR WALL MOUNTING
2. PROVIDE WITH REMOTE MOUNTING VFD.
3. PROVIDE WITH GRAVITY BACKDRAFT DAMPERS AT EACH FAN.
4. REFER TO CONTROL DIAGRAMS FOR ADDITIONAL CONTROL INSTRUCTIONS.

AIR HANDLING UNIT SCHEDULE

MARK AHU-	LOCATION	TYPE	CFM	O/A CFM	ESP S.P. IN WG	COOLING DATA: ENTERING DB	ENTERING WB	MIN CAP (MBH) SENS	MIN CAP (MBH) TOTAL	CHILLED WATER GPM	CHILLED WATER ENT	CHILLED WATER LVG	MAX PD (FT WG)	PRE-HEAT COIL: EAT (DEG F)	MIN CAP (MBH)	HEATING WATER GPM	HEATING WATER ENT	HEATING WATER LVG	MAX PD (FT WG)	FAN DATA: QTY	HP (EACH)	UNIT MCA	VOLTS	PH	WIEGHT	MANUFACTURER/ MODEL SERIES	REMARKS
1	PENTHOUSE	HORIZ DRAW THROUGH (E)	100,000	30,000	4.0	84.8	66.9	3341.8	3951.0	660.0	42.0	56.0	22.0	(E)	(E)	(E)	(E)	(E)	(E)	18	3.5	2@42.7	460	3	(E)	(E)	1,2,3,4,5,6,7
2	PENTHOUSE	HORIZ DRAW THROUGH (E)	100,000	30,000	4.0	84.8	66.9	3341.8	3951.0	660.0	42.0	56.0	22.0	(E)	(E)	(E)	(E)	(E)	(E)	18	3.5	2@42.7	460	3	(E)	(E)	1,2,3,4,5,6,7

REMARKS:
1. DOES NOT INCLUDE FILTER OR UNIT LOSSES
2. 4-SECTION COOLING COIL WITH BLANK OFF SECTIONS AS REQUIRED TO FIT EXISTING CONSTRUCTION. PROVIDE FACE DAMPERS ON THREE SECTIONS FOR CAPACITY CONTROL.
3. EXISTING UNIT TO BE PROVIDED WITH NEW FAN AND CHILLED WATER COIL SECTIONS
4. FAN SECTION SHALL BE TWO SEPARATE FAN WALLS WITH SEPARATE REMOTE MOUNTING MSP PANEL AND REMOTE MOUNTING VFD.
5. PROVIDE AUXILIARY DRAIN PAN SYSTEM
6. EXISTING PNEUMATIC CONTROLS TO BE REPLACED WITH DDC BY ENVIROMATIC (RELIABLE CONTROLS)
7. REFER TO CONTROL DIAGRAMS FOR ADDITIONAL INSTRUCTIONS.

Summit CONSULTANTS, INC.
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Texas BPE Registration # F-207 Exp. Date 6/30/12

STATE OF TEXAS 7/2/2018 RENE A. CULROSS 71739 LICENSED PROFESSIONAL ENGINEER
9438.7.10.2018.16184

Tarrant County Plaza Building Air Handler Replacement Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY, TEXAS

JOB NUMBER: P16184

DATE ISSUED: JULY 02, 2018

ISSUES:

ISSUE FOR REVIEW	03/24/2017
ISSUE FOR PRICING	03/31/2017
ISSUED FOR REPRICING	05/30/2018
ISSUE FOR BIDDING	07/02/2018

REVISIONS:

SHEET CONTENTS: MECHANICAL NOTES AND SCHEDULES

SHEET NUMBER: 1M001

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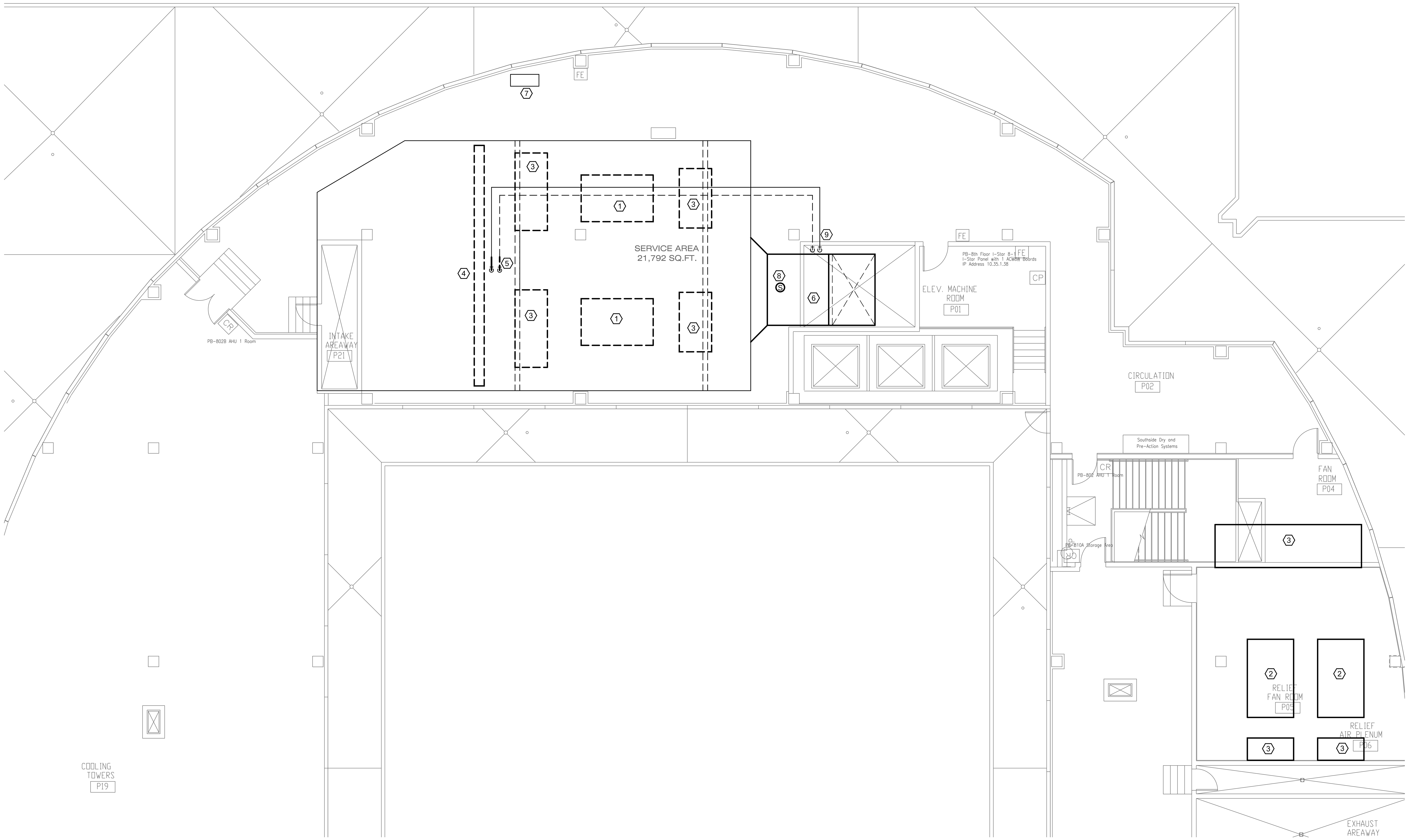


1 LEVEL 8 AHU-1 MECHANICAL ROOM DEMOLITION

SCALE: 1/8" = 1'-0"

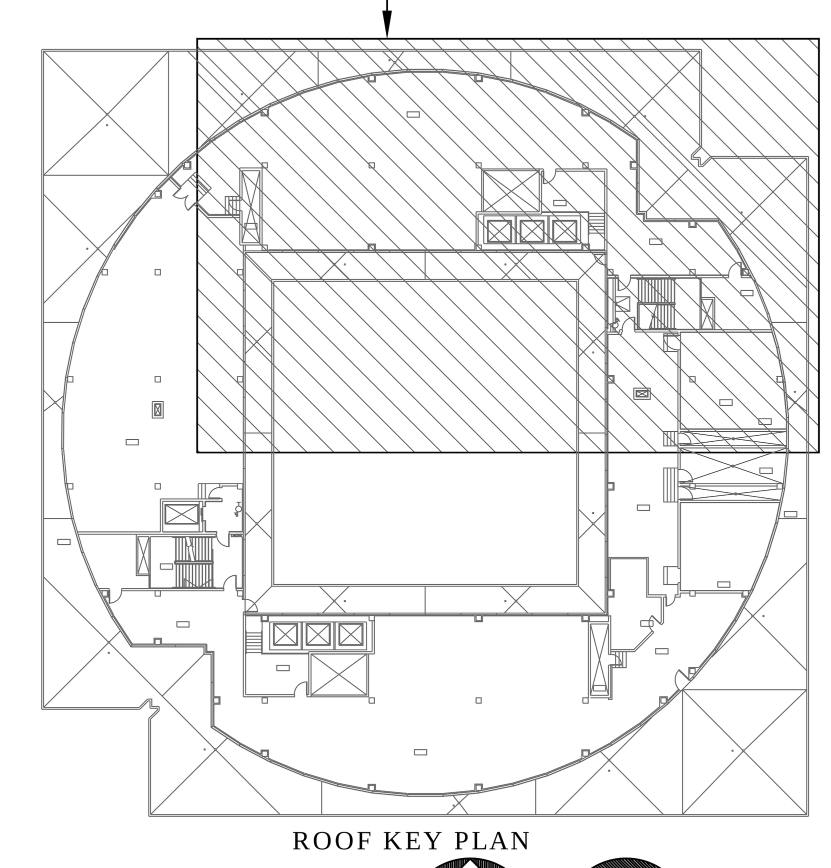
GENERAL NOTES

1. REFER TO SHEET 1M001 FOR SYMBOLS LEGENDS, AND ADDITIONAL NOTES.
2. WORK SHALL BE PHASED SUCH THAT ONLY ONE AIR HANDLING UNIT IS OUT OF SERVICE AT ANY ONE TIME. WORK ON SECOND UNIT MAY NOT COMMENCE UNTIL FIRST UNIT IS ON-LINE AND PROPER OPERATION IS CONFIRMED.
3. ELECTRICAL AND CONTROLS WORK MUST ALSO BE PHASED SO THAT ONLY ONE UNIT IS DEACTIVATED AT ANY ONE TIME.
4. SCHEDULE NECESSARY SHUTDOWNS OF CHILLED WATER SYSTEM TO BE DURING UNOCCUPIED HOURS. COORDINATE WITH FACILITIES TO ALLOW A MINIMUM OF ONE WEEK ADVANCE NOTICE OF THE SHUTDOWN.

NOTES BY SYMBOL "#"

1. DEMOLISH EXISTING SUPPLY AIR FAN AND ASSOCIATED, CONTROLS, WIRING, CONDUIT, AND ASSOCIATED FITTINGS AND ACCESSORIES. DEMOLISH ELECTRICAL CIRCUITS TO POINT OF CONNECTION AT PANEL. DEMOLISH INERTIA BASE AND EQUIPMENT PAD. RE-FINISH ENTIRE FLOOR IN SECTION WHERE EQUIPMENT PAD HAS BEEN REMOVED FOR A LEVEL AND SMOOTH FLOOR SURFACE. REFER TO DEMOLITION WORK NOTES ON 1M001 FOR ADDITIONAL INSTRUCTIONS INCLUDING DISPOSAL OF EXISTING EQUIPMENT.
2. DEMOLISH EXISTING RELIEF AIR FAN AND ASSOCIATED, CONTROLS, WIRING, CONDUIT, AND ASSOCIATED FITTINGS AND ACCESSORIES. DEMOLISH ELECTRICAL CIRCUITS TO POINT OF CONNECTION AT PANEL. DEMOLISH INERTIA BASE AND EQUIPMENT PAD. RE-FINISH ENTIRE FLOOR IN SECTION WHERE EQUIPMENT PAD HAS BEEN REMOVED FOR A LEVEL AND SMOOTH FLOOR SURFACE. REFER TO DEMOLITION WORK NOTES ON 1M001 FOR ADDITIONAL INSTRUCTIONS INCLUDING DISPOSAL OF EXISTING EQUIPMENT.
3. DEMOLISH EXISTING SOUND ATTENUATOR AND ASSOCIATED FITTINGS AND ACCESSORIES.
4. DEMOLISH EXISTING COOLING COIL, CONNECTING PIPING AND ASSOCIATED FITTINGS AND ACCESSORIES.
5. DEMOLISH EXISTING CHILLED WATER PIPING AS SHOWN. TEMPORARILY CAP OPEN PIPING WITH SAME MATERIAL AS EXISTING.
6. DEMOLISH ACOUSTICALLY LINED SUPPLY DUCTWORK BETWEEN AHU DISCHARGE AND POINT OF TRANSITION TO EXTERNALLY LINED DUCTWORK (APPROXIMATELY 50-FEET). DEMOLITION WILL INCLUDE THE SECURITY BARS WITHIN THE DUCTWORK, WHICH WILL BE REPLACED WITH SMACNA COMPLIANT INTERNAL DUCT SUPPORTS
7. APPROXIMATE LOCATION OF AHU CONTROLS PANEL. RETAIN FUNCTIONALITY OF CONTROLS UNTIL PARALLEL PANEL IS IN PLACE. CROSS-OVER TO NEW PANEL SHALL NOT OCCUR UNTIL AFTER UPGRADE WORK ON AHU IS COMPLETE.
8. DEMOLISH SMOKE DETECTOR AS PART OF DEMOLITION OF SUPPLY DUCTWORK. COORDINATE DE-ACTIVATION OF THIS DEVICE WITH FIRE ALARM INSTALLER TO PREVENT NUISANCE ALARMS.
9. 8-INCH VALVE ASSEMBLY FOR BYPASS OPERATION TO BE REPLACED. 3-WAY VALVE OPERATION AT COIL WILL CONTINUE UNTIL PHASE 2 IS COMPLETE. (NOTE (4) VALVES TOTAL EACH PENTHOUSE).

AREA OF WORK



ROOF KEY PLAN

8' 0 8' 16'

1/8"=1'-0"

PLAN TRUE

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Texas BPE Registration # F-207 Exp. Date 6

STATE OF TEXAS 7/2/2018 RENÉ A. CULROSS 71739 LICENSED PROFESSIONAL ENGINEER

9438.7.10.2018.16184

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TARRANT COUNTY, TEXAS

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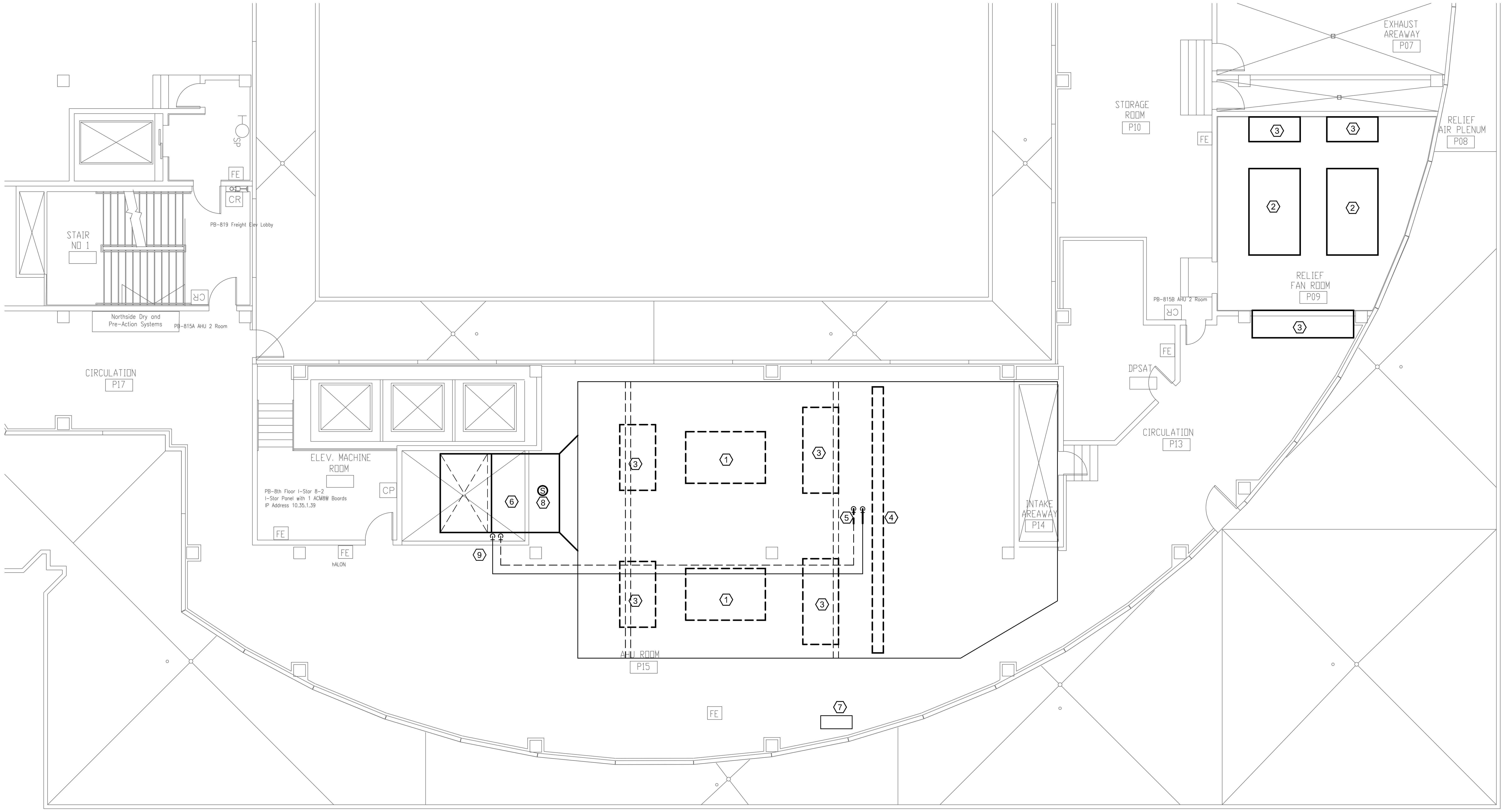
ISSUE FOR BIDDING	07/02

REVISIONS:

SHEET CONTENTS:
LEVEL 8
AHU-1 MECH R
DEMOLITION

SHEET NUMBER:

1M108A



1 LEVEL 8 AHU-2 MECHANICAL ROOM DEMOLITION

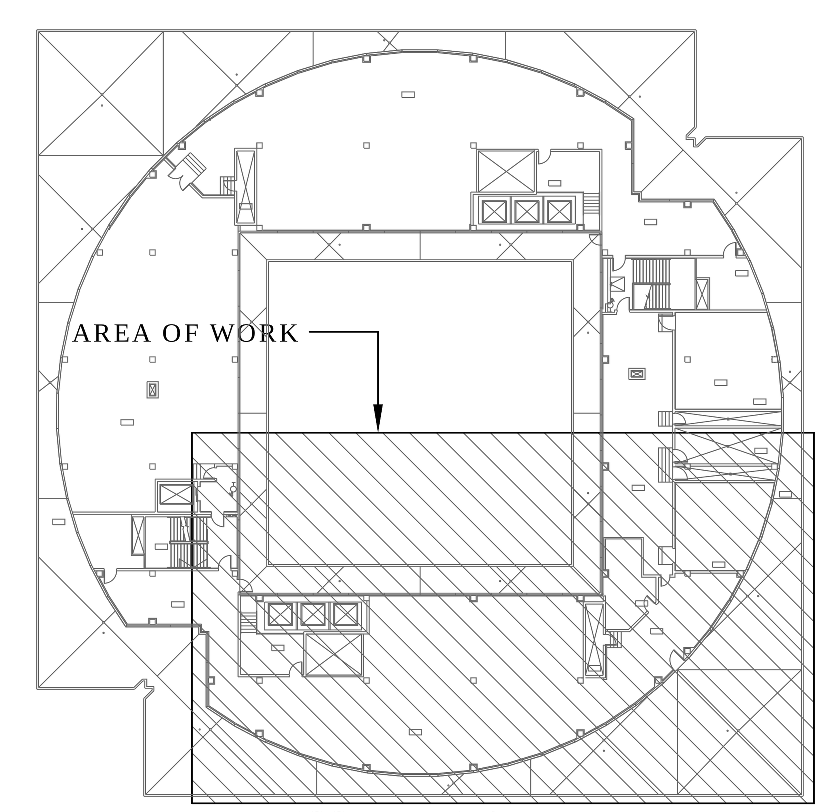
SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. REFER TO SHEET 1M001 FOR SYMBOLS LEGENDS, AND ADDITIONAL NOTES.
2. WORK SHALL BE PHASED SUCH THAT ONLY ONE AIR HANDLING UNIT IS OUT OF SERVICE AT ANY ONE TIME. WORK ON SECOND UNIT MAY NOT COMMENCE UNTIL FIRST UNIT IS ON-LINE AND PROPER OPERATION IS CONFIRMED.
3. ELECTRICAL AND CONTROLS WORK MUST ALSO BE PHASED SO THAT ONLY ONE UNIT IS DEACTIVATED AT ANY ONE TIME.
4. SCHEDULE NECESSARY SHUTDOWNS OF CHILLED WATER SYSTEM TO BE DURING UNOCCUPIED HOURS. COORDINATE WITH FACILITIES TO ALLOW A MINIMUM OF ONE WEEK ADVANCE NOTICE OF THE SHUTDOWN.

NOTES BY SYMBOL "#"

1. DEMOLISH EXISTING SUPPLY AIR FAN AND ASSOCIATED, CONTROLS, WIRING, CONDUIT, AND ASSOCIATED FITTINGS AND ACCESSORIES. DEMOLISH ELECTRICAL CIRCUITS TO POINT OF CONNECTION AT PANEL. DEMOLISH INERTIA BASE AND EQUIPMENT PAD. RE-FINISH ENTIRE FLOOR IN SECTION WHERE EQUIPMENT PAD HAS BEEN REMOVED FOR A LEVEL AND SMOOTH FLOOR SURFACE. REFER TO DEMOLITION WORK NOTES ON 1M001 FOR ADDITIONAL INSTRUCTIONS INCLUDING DISPOSAL OF EXISTING EQUIPMENT.
2. DEMOLISH EXISTING RELIEF AIR FAN AND ASSOCIATED, CONTROLS, WIRING, CONDUIT, AND ASSOCIATED FITTINGS AND ACCESSORIES. DEMOLISH ELECTRICAL CIRCUITS TO POINT OF CONNECTION AT PANEL. DEMOLISH INERTIA BASE AND EQUIPMENT PAD. RE-FINISH ENTIRE FLOOR IN SECTION WHERE EQUIPMENT PAD HAS BEEN REMOVED FOR A LEVEL AND SMOOTH FLOOR SURFACE. REFER TO DEMOLITION WORK NOTES ON 1M001 FOR ADDITIONAL INSTRUCTIONS INCLUDING DISPOSAL OF EXISTING EQUIPMENT.
3. DEMOLISH EXISTING SOUND ATTENUATOR AND ASSOCIATED FITTINGS AND ACCESSORIES.
4. DEMOLISH EXISTING COOLING COIL, CONNECTING PIPING AND ASSOCIATED FITTINGS AND ACCESSORIES.
5. DEMOLISH EXISTING CHILLED WATER PIPING AS SHOWN. TEMPORARILY CAP OPEN PIPING WITH SAME MATERIAL AS EXISTING.
6. DEMOLISH ACOUSTICALLY LINED SUPPLY DUCTWORK BETWEEN AHU DISCHARGE AND POINT OF TRANSITION TO EXTERNALLY LINED DUCTWORK (APPROXIMATELY 50-FEET). DEMOLITION WILL INCLUDE THE SECURITY BARS WITHIN THE DUCTWORK, WHICH WILL BE REPLACED WITH SMACNA COMPLIANT INTERNAL DUCT SUPPORTS
7. APPROXIMATE LOCATION OF AHU CONTROLS PANEL. RETAIN FUNCTIONALITY OF CONTROLS UNTIL PARALLEL PANEL IS IN PLACE. CROSS-OVER TO NEW PANEL SHALL NOT OCCUR UNTIL AFTER UPGRADE WORK ON AHU IS COMPLETE.
8. DEMOLISH SMOKE DETECTOR AS PART OF DEMOLITION OF SUPPLY DUCTWORK. COORDINATE DE-ACTIVATION OF THIS DEVICE WITH FIRE ALARM INSTALLER TO PREVENT NUISANCE ALARMS.
9. 8-INCH VALVE ASSEMBLY FOR BYPASS OPERATION TO BE REPLACED. 3-WAY VALVE OPERATION AT COIL WILL CONTINUE UNTIL PHASE 2 IS COMPLETE. (NOTE (4) VALVES TOTAL EACH PENTHOUSE).



ROOF KEY PLAN

8' 0 8' 16'

1/8"=1'-0"

PLAN TRUE

Summit CONSULTANTS, INC.
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STATE OF TEXAS 7/2/2018 RENE A. CULROSS 71739 LICENSED PROFESSIONAL ENGINEER
9438.7.10.2018.16184

Tarrant County Plaza Building Air Handler Replacement Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY TEXAS

JOB NUMBER: P16184

DATE ISSUED: JULY 02, 2018

ISSUES:

ISSUE FOR REVIEW	03/24/2017
ISSUE FOR PRICING	03/31/2017
ISSUED FOR REPRICING	05/30/2018
ISSUE FOR BIDDING	07/02/2018

REVISIONS:

SHEET CONTENTS:
LEVEL 8
AHU-2 MECH RM
DEMOLITION

SHEET NUMBER:

1M108B

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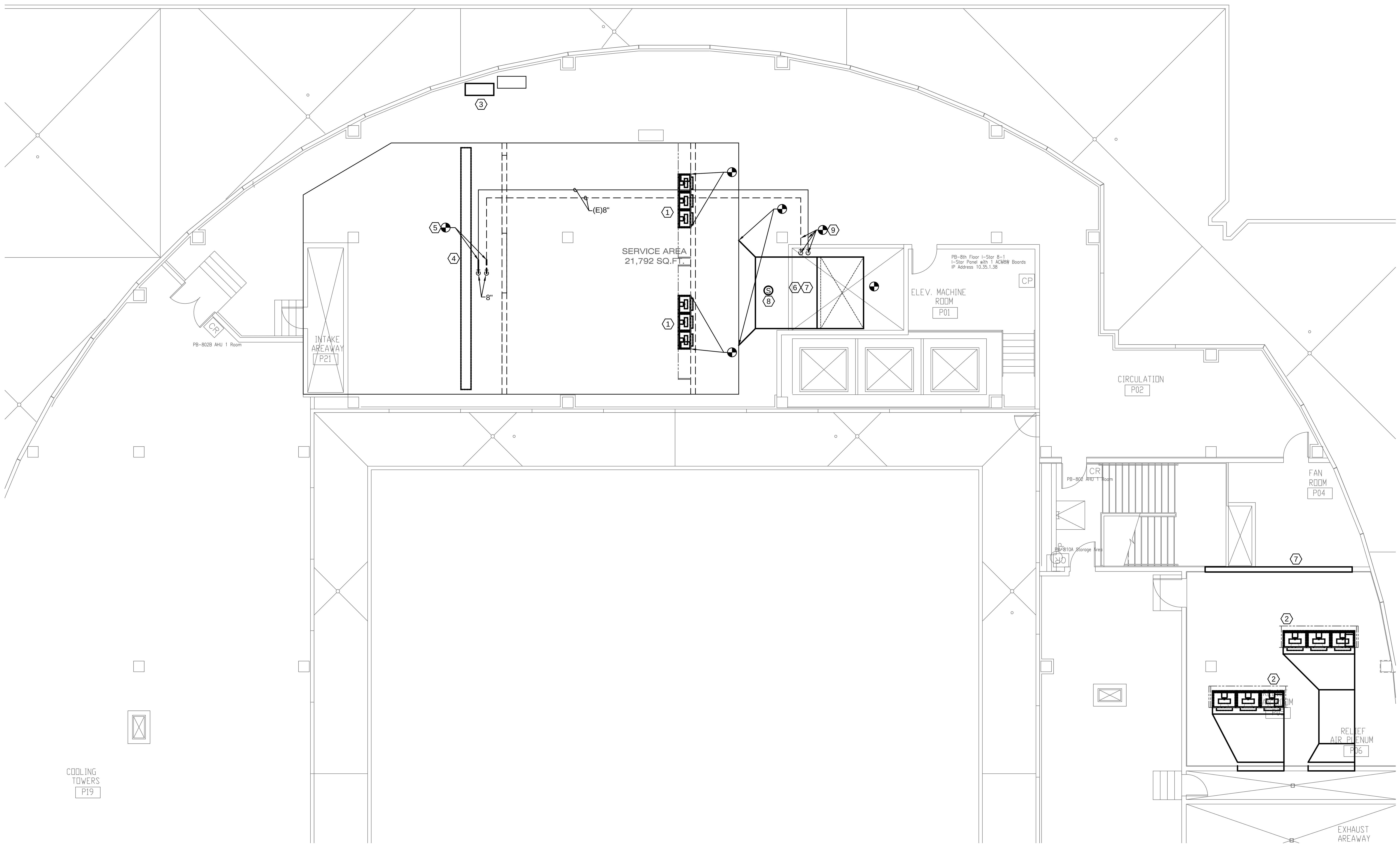


1 LEVEL 8 AHU-1 MECHANICAL ROOM RENOVATION
SCALE: 1/8" = 1'-0"

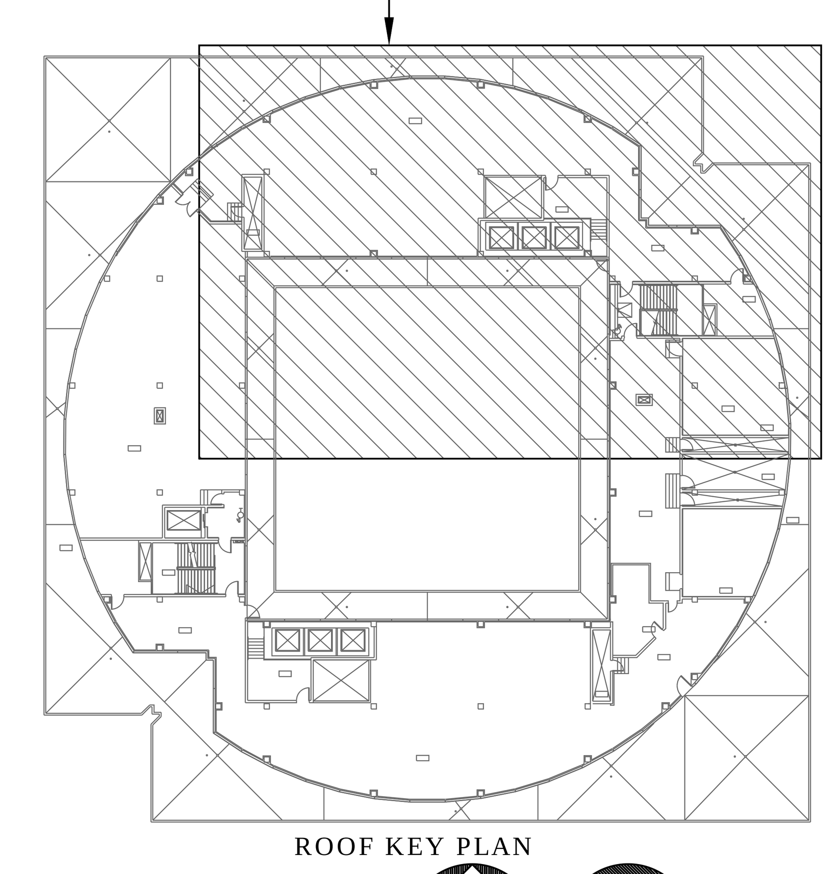
GENERAL NOTES

1. REFER TO SHEET 1M001 FOR SYMBOLS LEGENDS, AND ADDITIONAL NOTES.
2. WORK SHALL BE PHASED SUCH THAT ONLY ONE AIR HANDLING UNIT IS OUT OF SERVICE AT ANY ONE TIME. WORK ON SECOND UNIT MAY NOT COMMENCE UNTIL FIRST UNIT IS ON-LINE AND PROPER OPERATION IS CONFIRMED.
3. ELECTRICAL AND CONTROLS WORK MUST ALSO BE PHASED SO THAT ONLY ONE UNIT IS DEACTIVATED AT ANY ONE TIME.
4. SCHEDULE NECESSARY SHUTDOWNS OF CHILLED WATER SYSTEM TO BE DURING UNOCCUPIED HOURS. COORDINATE WITH FACILITIES TO ALLOW A MINIMUM OF ONE WEEK ADVANCE NOTICE OF THE SHUTDOWN.

NOTES BY SYMBOL "#"

1. FAN WALL REPLACEMENT SUPPLY FANS WITH VFD CONTROL. REFER TO CONTROL DIAGRAMS ON 1M501 FOR ADDITIONAL INSTRUCTIONS.
2. FAN WALL REPLACEMENT RELIEF FANS WITH VFD CONTROL. REFER TO CONTROL DIAGRAMS ON 1M501 FOR ADDITIONAL INSTRUCTIONS.
3. PROPOSED LOCATION OF NEW CONTROL PANEL. COORDINATE FINAL LOCATION WITH OTHER EQUIPMENT. DEMOLISH EXISTING CABINET ONCE CONTROLS FOR REPLACEMENT AIR HANDLER ARE OPERATIONAL.
4. REPLACEMENT COOLING COIL. REFER TO AIR HANDLING UNIT SCHEDULE FOR CAPACITY REQUIREMENTS. PROVIDE INSULATED BLANK-OFF SECTION TO ACCOMMODATE REDUCED SIZE COIL.
5. RE-CCONNECT CHILLED WATER PIPING WHERE EXISTING WAS REMOVED. EXTEND 8"Ø PIPING DOWN TO COIL HEADERS AT 4-SECTION COILS.
6. TRANSITION FROM FULL OPENING SIZE OF AHU DISCHARGE TO 100/72 (FIELD VERIFY). EXTEND DOWN IN CHASE TO POINT OF CONNECTION WHERE EXISTING WAS REMOVED (APPROXIMATELY 50-FEET). DUCTWORK TO BE EXTERNALLY WRAPPED WITH RIGID OR BLANKET INSULATION MEETING AND INSTALLED R-VALUE AS REQUIRED BY IECC. PROVIDE INTERNAL DUCT SUPPORTS IN ACCORDANCE WITH SMACNA GUIDELINES.
7. REPAIR AND FRAME OPENING WHERE SOUND ATTENUATOR WAS REMOVED.
8. NEW SMOKE DETECTOR TO BE PROVIDED BY MECHANICAL SUB-CONTRACTOR AND INSTALLED/WIRED BY STATE CERTIFIED FIRE ALARM SYSTEM INSTALLER.
9. INSTALL (4) 8-INCH ISOLATION VALVES AND BYPASS CONTROL VALVES FOR TEMPORARY 3-WAY VALVE OPERATION AT THE COIL. ONCE PHASE 2 IS COMPLETE, COIL VALVES WILL BE CONVERTED TO 2-WAY COIL OPERATION, BUT BYPASS AT MAIN WILL BE ADJUSTED TO MINIMUM FLOW CONTROL BYPASS. PROVIDE NEW TEMPERATURE AND PRESSURE GAUGES AND SENSORS FOR INPUT TO BUILDING MANAGEMENT SYSTEM.

AREA OF WORK



ROOF KEY PLAN

8' 0 8' 16'
1/8"=1'-0"

N PLAN N TRUE

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STATE OF TEXAS
7/2/2018
RENE A. CULROSS
71739
LICENSED
PROFESSIONAL ENGINEER
9438.7.10.2018.16184

Tarrant County Plaza Building Air Handler Replacement Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY, TEXAS

JOB NUMBER: P16184

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ISSUE FOR PRICING	03/31/2017
ISSUED FOR REPRICING	05/30/2018
ISSUE FOR BIDDING	07/02/2018

REVISIONS:

SHEET CONTENTS:
LEVEL 8
AHU-1 MECH RM
RENOVATION

SHEET NUMBER:
1M208A

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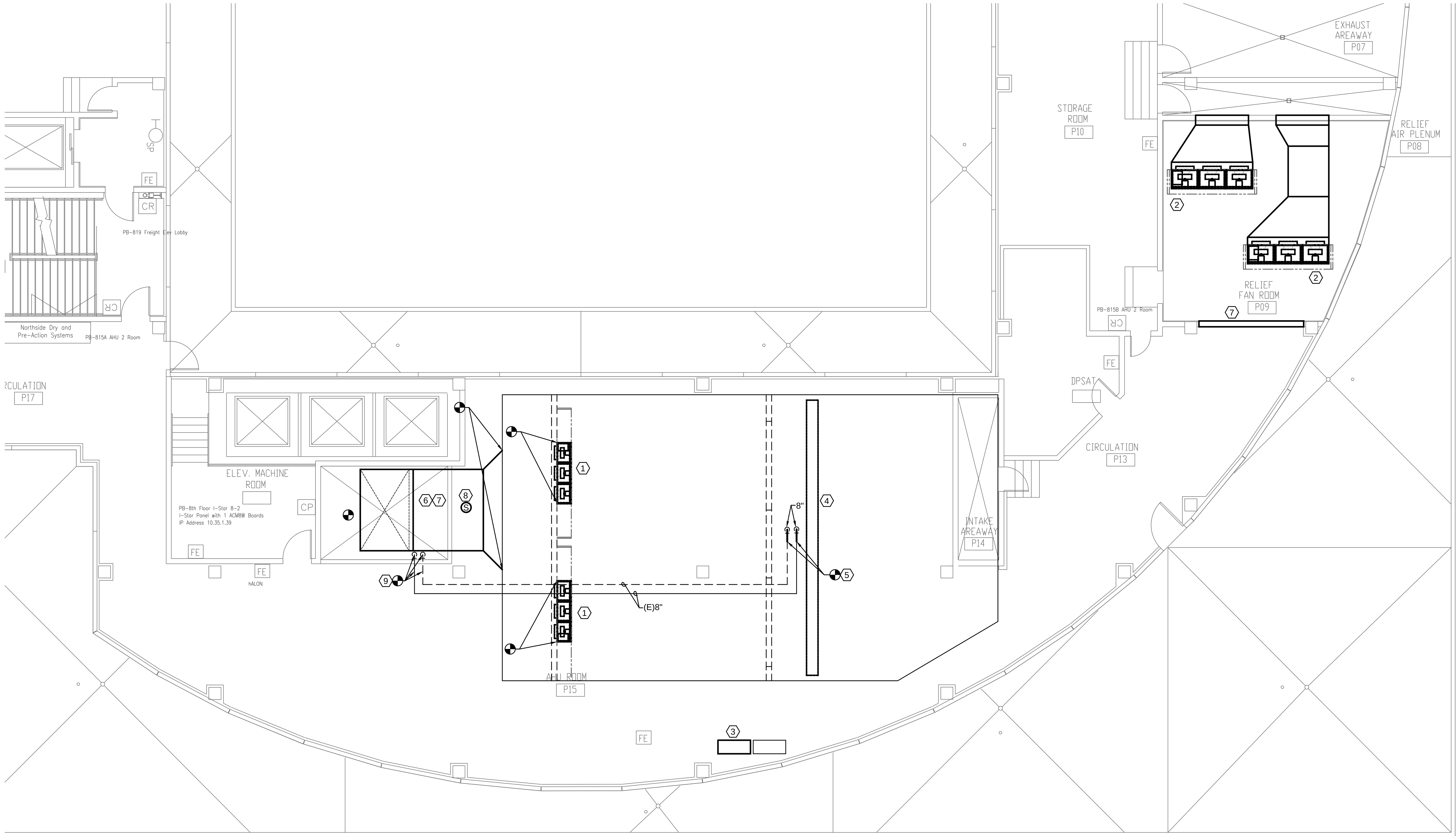


1 LEVEL 8 AHU-2 MECHANICAL ROOM RENOVATION
SCALE: 1/8" = 1'-0"

GENERAL NOTES

1. REFER TO SHEET 1M001 FOR SYMBOLS LEGENDS, AND ADDITIONAL NOTES.
2. WORK SHALL BE PHASED SUCH THAT ONLY ONE AIR HANDLING UNIT IS OUT OF SERVICE AT ANY ONE TIME. WORK ON SECOND UNIT MAY NOT COMMENCE UNTIL FIRST UNIT IS ON-LINE AND PROPER OPERATION IS CONFIRMED.
3. ELECTRICAL AND CONTROLS WORK MUST ALSO BE PHASED SO THAT ONLY ONE UNIT IS DEACTIVATED AT ANY ONE TIME.
4. SCHEDULE NECESSARY SHUTDOWNS OF CHILLED WATER SYSTEM TO BE DURING UNOCCUPIED HOURS. COORDINATE WITH FACILITIES TO ALLOW A MINIMUM OF ONE WEEK ADVANCE NOTICE OF THE SHUTDOWN.

NOTES BY SYMBOL "#"

1 FAN WALL REPLACEMENT SUPPLY FANS WITH VFD CONTROL. REFER TO CONTROL DIAGRAMS ON 1M501 FOR ADDITIONAL INSTRUCTIONS.

2 FAN WALL REPLACEMENT RELIEF FANS WITH VFD CONTROL. REFER TO CONTROL DIAGRAMS ON 1M501 FOR ADDITIONAL INSTRUCTIONS.

3 PROPOSED LOCATION OF NEW CONTROL PANEL. COORDINATE FINAL LOCATION WITH OTHER EQUIPMENT. DEMOLISH EXISTING CABINET ONCE CONTROLS FOR REPLACEMENT AIR HANDLER ARE OPERATIONAL.

4 REPLACEMENT COOLING COIL. REFER TO AIR HANDLING UNIT SCHEDULE FOR CAPACITY REQUIREMENTS. PROVIDE INSULATED BLANK-OFF SECTION TO ACCOMMODATE REDUCED SIZE COIL.

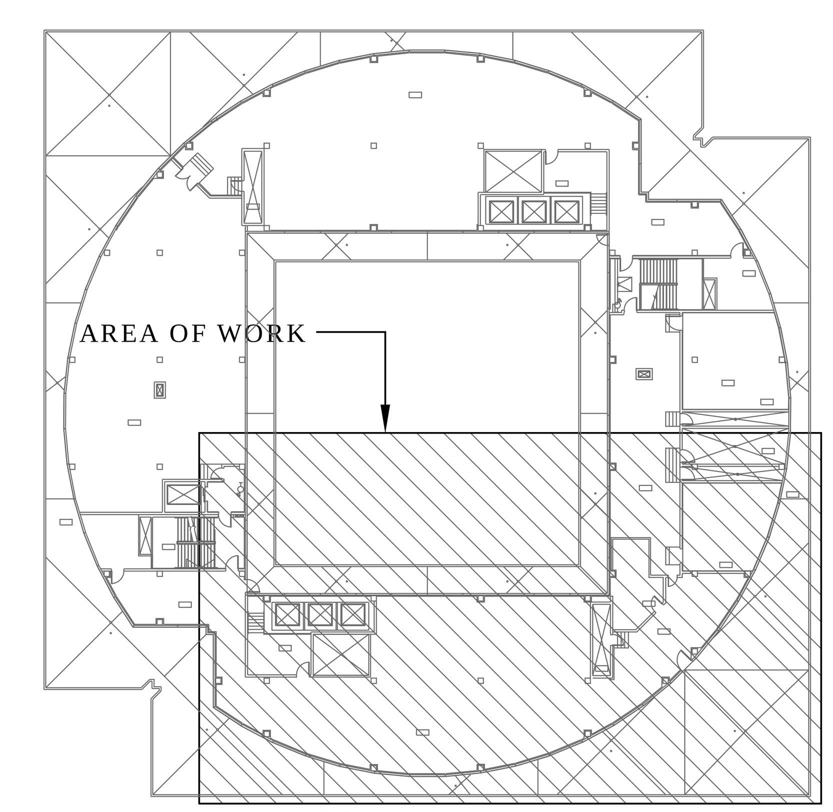
5 RE-CCONNECT CHILLED WATER PIPING WHERE EXISTING WAS REMOVED. EXTEND 8"Ø PIPING DOWN TO COIL HEADERS AT 4-SECTION COILS.

6 TRANSITION FROM FULL OPENING SIZE OF AHU DISCHARGE TO 100/72 (FIELD VERIFY). EXTEND DOWN IN CHASE TO POINT OF CONNECTION WHERE EXISTING WAS REMOVED (APPROXIMATELY 50-FEET). DUCTWORK TO BE EXTERNALLY WRAPPED WITH RIGID OR BLANKET INSULATION MEETING AND INSTALLED R-VALUE AS REQUIRED BY IECC. PROVIDE INTERNAL DUCT SUPPORTS IN ACCORDANCE WITH SMACNA GUIDELINES.

7 REPAIR AND FRAME OPENING WHERE SOUND ATTENUATOR WAS REMOVED.

8 NEW SMOKE DETECTOR TO BE PROVIDED BY MECHANICAL SUB-CONTRACTOR AND INSTALLED/WIRED BY STATE CERTIFIED FIRE ALARM SYSTEM INSTALLER.

9 INSTALL (4) 8-INCH ISOLATION VALVES AND BYPASS CONTROL VALVES FOR TEMPORARY 3-WAY VALVE OPERATION AT THE COIL. ONCE PHASE 2 IS COMPLETE, COIL VALVES WILL BE CONVERTED TO 2-WAY COIL OPERATION, BUT BYPASS AT MAIN WILL BE ADJUSTED TO MINIMUM FLOW CONTROL BYPASS. PROVIDE NEW TEMPERATURE AND PRESSURE GAUGES AND SENSORS FOR INPUT TO BUILDING MANAGEMENT SYSTEM.



ROOF KEY PLAN

8' 0 8' 16'
1/8"=1'-0"

PLAN TRUE

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STATE OF TEXAS
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RENÉ A. CULROSS
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9438.7.10.2018.16184

Tarrant County Plaza Building Air Handler Replacement Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY TEXAS

JOB NUMBER:
P16184

DATE ISSUED:
JULY 02, 2018

ISSUES:

ISSUE FOR REVIEW	03/24/2017
ISSUE FOR PRICING	03/31/2017
ISSUED FOR REPRICING	05/30/2018
ISSUE FOR BIDDING	07/02/2018

REVISIONS:

SHEET CONTENTS:
LEVEL 8
AHU-2 MECH RM
RENOVATION

SHEET NUMBER:
1M208B

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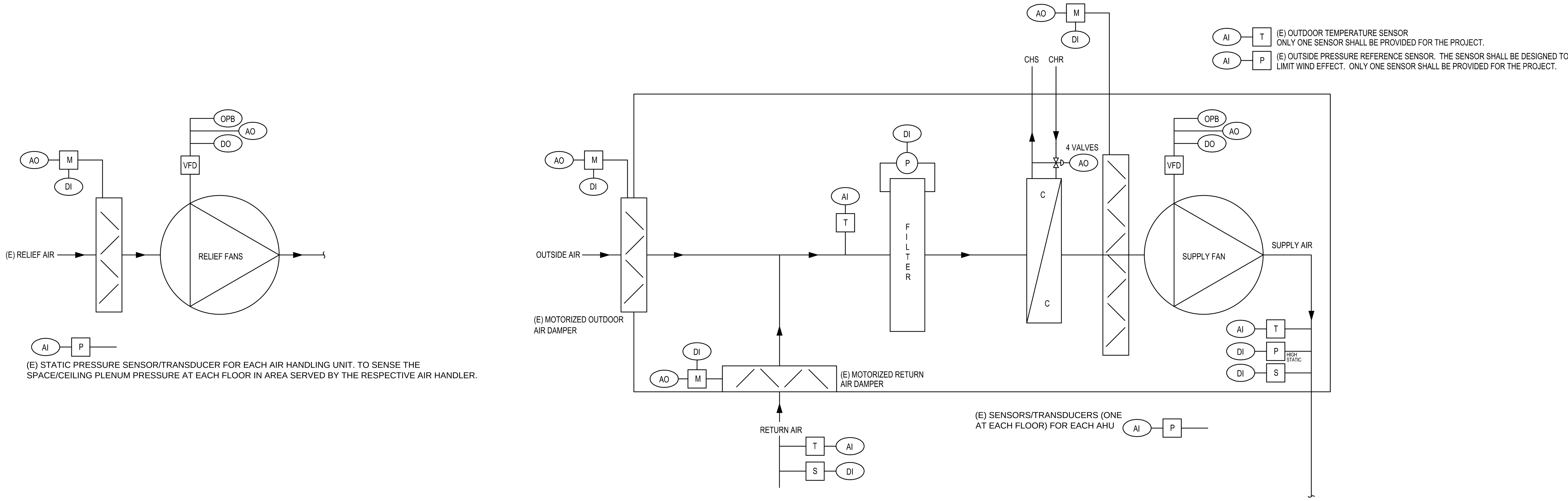

1 AIR HANDLING SYSTEM CONTROL DIAGRAM

NO SCALE

SEQUENCE OF OPERATION - AHU-1, 2

1. EXISTING SEQUENCE OF OPERATION IS TO BE MAINTAINED UNTIL PHASE 2 IS IMPLEMENTED WITH EXCEPTIONS NOTED BELOW.
2. EACH FAN IN EACH ARRAY SHALL BE INDIVIDUALLY CONTROLLABLE FOR ON/OFF FUNCTION FROM THE MAIN CONTROL SYSTEM TERMINAL AT THE COUNTY FACILITIES OFFICE.
3. EACH FAN IN EACH ARRAY SHALL BE PROVIDED WITH MONITORING CAPABILITY FOR STATUS (ON/OFF) AND STATIC PRESSURE.
4. UNOCCUPIED MODE: SUPPLY FANS ARE OFF.
5. OCCUPIED MODE: SUPPLY FANS OPERATING ON VFD OPERATION.
6. 4-SECTION COOLING COIL VALVES MODULATE IN SEQUENCE TO MAINTAIN SUPPLY AIR TEMPERATURE SET POINT, WITH THE OUTER COILS ACTIVATING ABOVE 50 AND 75-PERCENT CAPACITY RESPECTIVELY. INNER COILS MODULATE AT BELOW 50- AND 25-PERCENT CAPACITY RESPECTIVELY. BOTH INNER COILS ARE AT 100-PERCENT OPEN AT 50-PERCENT CAPACITY. WHEN OUTER COILS ARE INACTIVE (MODULATING VALVE FULLY CLOSED TO THE COIL), THE FACE DAMPERS SHALL ALSO CLOSE.
7. REFER TO 1M208A AND 1M208B FOR LOCATION OF CHILLED WATER PRESSURE AND TEMPERATURE SENSORS.
8. RELIEF AIR FANS ARE TO BE SEQUENCED TO MAINTAIN BUILDING PRESSURE AT 0.05-INCHES STATIC PRESSURE (ADJUSTABLE) WITH A MINIMUM SETTING OF SCHEDULED OUTDOOR AIR VALUE.
9. SAFETY SHUTDOWN OF THE FANS. THE CONTROL SYSTEM SHUTS DOWN THE FANS IF THERE IS A LOW TEMPERATURE CONDITION, IF SMOKE IS DETECTED, OR IF A HIGH STATIC PRESSURE CONDITION IS DETECTED.
   a. LOW COOLING COIL TEMPERATURE DETECTION. ON A FALL IN TEMPERATURE AT THE SENSOR LOCATED UPSTREAM OF THE COOLING COIL TO ITS SETPOINT (35°), THE FANS SHALL BE SHUTDOWN. AND THE OUTSIDE AIR DAMPER CLOSED. THE LOW TEMPERATURE PROTECTION INDICATOR AT THE CONTROL PANEL MUST BE MANUALLY
   b. SMOKE DETECTION. DUCT SMOKE DETECTORS STOP THE SUPPLY/RETURN FANS AND CLOSE THE UNIT ISOLATING DAMPERS WHENEVER THE PRESENCE OF SMOKE IS DETECTED. TO RESTART THE FANS, THE SMOKE DETECTORS AND THE CONTROL PANEL MUST BE MANUALLY RESET. THE OUTSIDE AIR DAMPER CLOSE.
   c. HIGH STATIC PRESSURE DETECTION. ON A RISE IN STATIC PRESSURE ABOVE THE SETPOINT OF A HIGH LIMIT STATIC PRESSURE SWITCH DOWNSTREAM OF THE SUPPLY FAN, THE FANS STOP. TO RESTART THE FANS, THE CONTROL PANEL MUST BE MANUALLY RESET.

CONTROL SYSTEM GENERAL NOTES

THE CONTROL SYSTEMS SHALL BE COMPLETE WITH ALL WIRING, CONDUIT, POWER SUPPLIES AND ALL OTHER ITEMS REQUIRED FOR A COMPLETE AND OPERATIONAL SYSTEM THAT WILL ACCOMPLISH THE SEQUENCE OF OPERATIONS.

COORDINATE ALL CONTROLS WORK WITH RELIABLE CONTROLS (CONTACT: SID ELLIS, sellis@enviromaticsystems.com, (972)206-2635)

IT IS THE CONTRACTOR'S RESPONSIBILITY TO COORDINATE ALL ASPECTS OF THE DDC CONTROL SYSTEM AND THE FIRE ALARM/SUPRESSION SYSTEMS TO ENSURE THAT THE SYSTEMS OPERATE AS REQUIRED BY THESE DOCUMENTS AND NATIONAL AND LOCAL CODES.

SYMBOL LIST

SYMBOL	DESCRIPTION
	OPPOSED BLADE DAMPER
	HEATING OR COOLING COIL
	AUTOMATIC 2-WAY VALVE
	AUTOMATIC 3-WAY VALVE
	FAN OR PUMP MOTOR
P	DIFFERENTIAL PRESSURE SENSOR
P	PRESSURE TRANSMITTER
CSR	CURRENT SENSING RELAY
FS	WATER FLOW MONITORING
S	SMOKE DETECTOR
T	TEMPERATURE SENSOR
T	THERMOSTAT OR TEMP SENSOR
H	HUMIDITSTAT OR HUMIDITY SENSOR
TCU	TERMINAL CONTROL UNIT
VFD	VARIABLE FREQUENCY DRIVE
VS	VIBRATION SENSOR
	VAV DAMPER W/FLOW MONITOR
DI	DDC DIGITAL INPUT POINT
DO	DDC DIGITAL OUTPUT POINT
AI	DDC ANALOG INPUT POINT
AO	DDC ANALOG OUTPUT POINT
OPB	OPEN PROTOCOL BUS
M	MOTOR
MS	MOTOR STARTER
ES	ENTHALPY SENSOR
HPL	HIGH PRESSURE LIMIT SWITCH
CO_2	CARBON DIOXIDE DETECTOR
F	AIRFLOW MONITORING STATION
HLH	HIGH LIMIT HUMIDITY
H	HUMIDITY SENSOR

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Tarrant County Plaza Building Air Handler Replacement Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY TEXAS

JOB NUMBER: P16184

DATE ISSUED: MARCH 31, 2017

ISSUES:

ISSUE FOR REVIEW	03/24/17
ISSUE FOR 100% REVIEW	
ISSUE FOR CONSTRUCTION	04/01/17
RECORD DRAWINGS	

REVISIONS:

SHEET CONTENTS: MECHANICAL CONTROLS

SHEET NUMBER: 1M501

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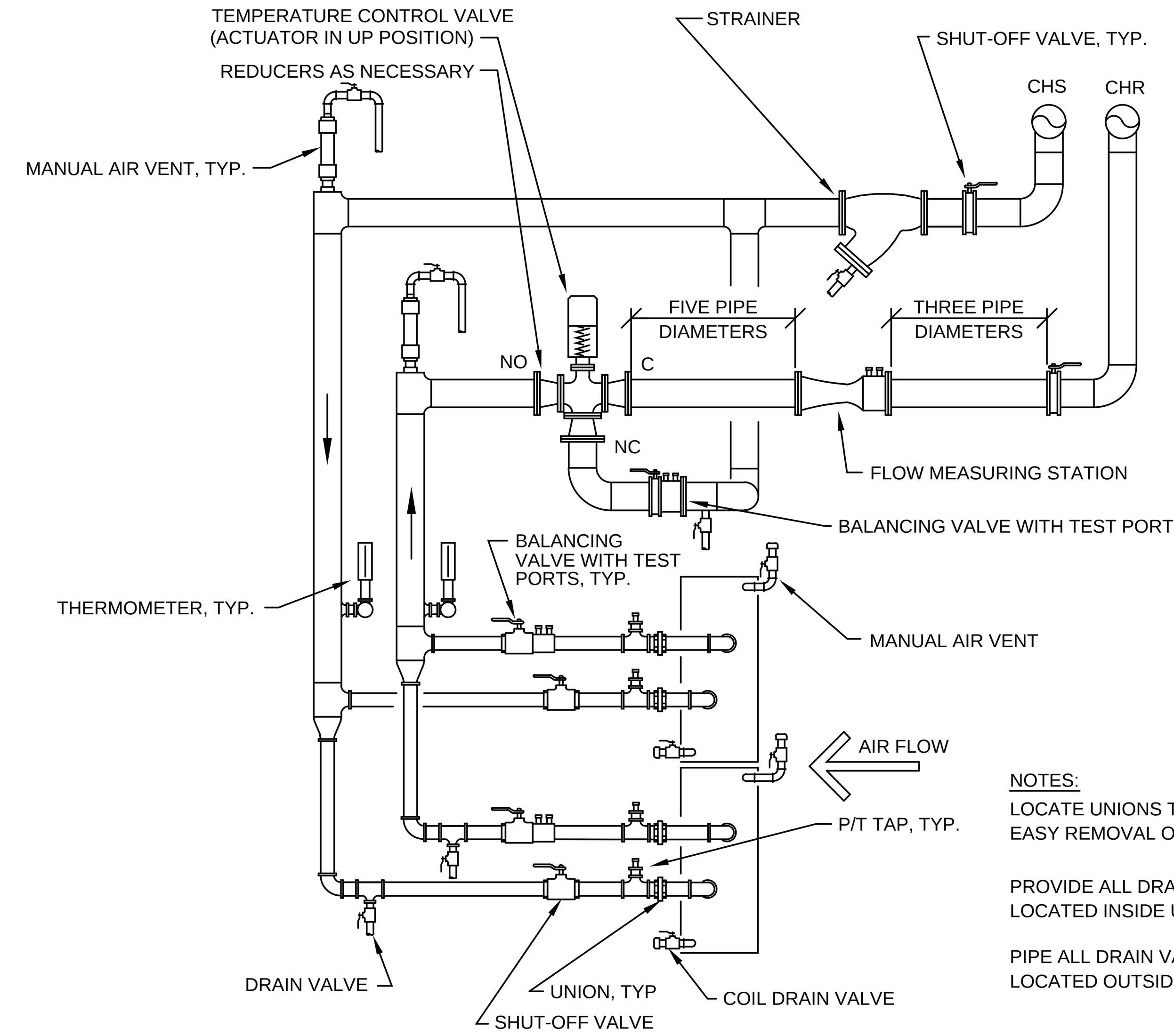


VALVE SCHEDULE NOTES:

SHUT-OFF VALVES:
THROUGH 2-1/2": BALL VALVE
ABOVE 2-1/2": BUTTERFLY VALVE

BALANCING VALVES:
THROUGH 2-1/2": BALL VALVE
ABOVE 2-1/2": BUTTERFLY VALVE
WITH P/T TAPS AND MEMORY STOP

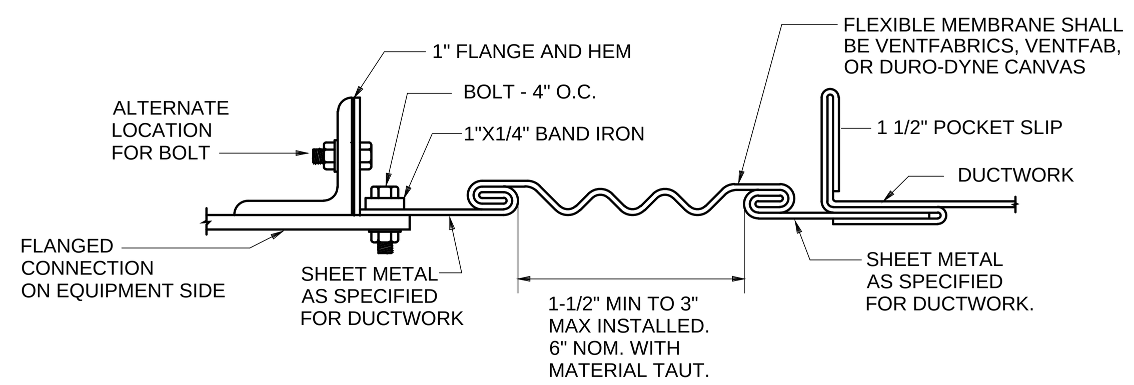
MANUAL AIR VENT:
BALL VALVE WITH HOSE END AND CAP

DRAIN VALVE:
BALL VALVE WITH HOSE END AND CAP

NOTES:
LOCATE UNIONS TO PROVIDE EASY REMOVAL OF COILS

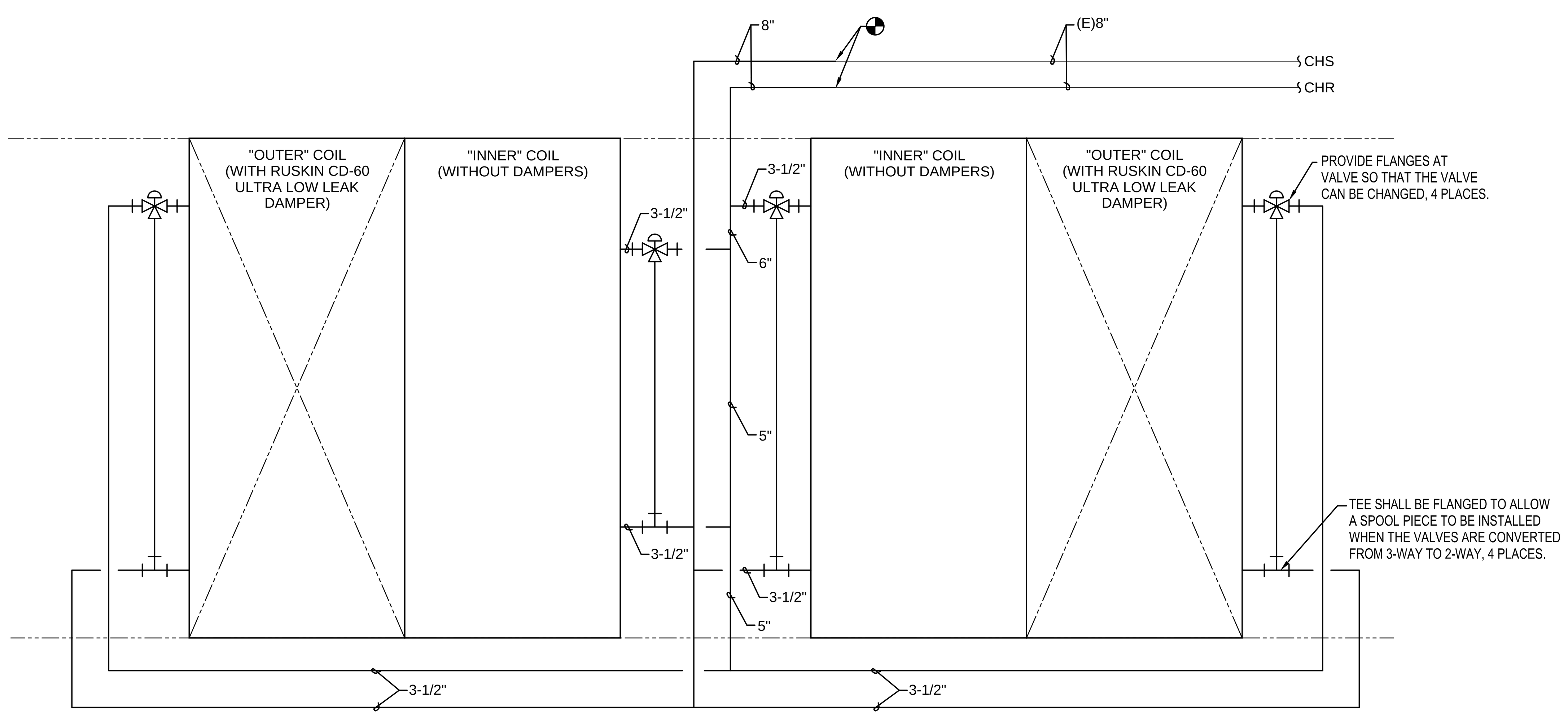
PROVIDE ALL DRAIN VALVES AND AIR VENTS LOCATED INSIDE UNIT WITH HOSE END AND CAP

PIPE ALL DRAIN VALVES AND AIR VENTS LOCATED OUTSIDE UNIT TO FLOOR DRAIN



2 FLEXIBLE EQUIPMENT CONNECTION DETAIL
SCALE: NO SCALE

1 MULTIPLE COOLING COILS WITH THREE-WAY VALVE
SCALE: NO SCALE



5 AHU-1 & 2 CHILLED WATER PIPING DIAGRAM
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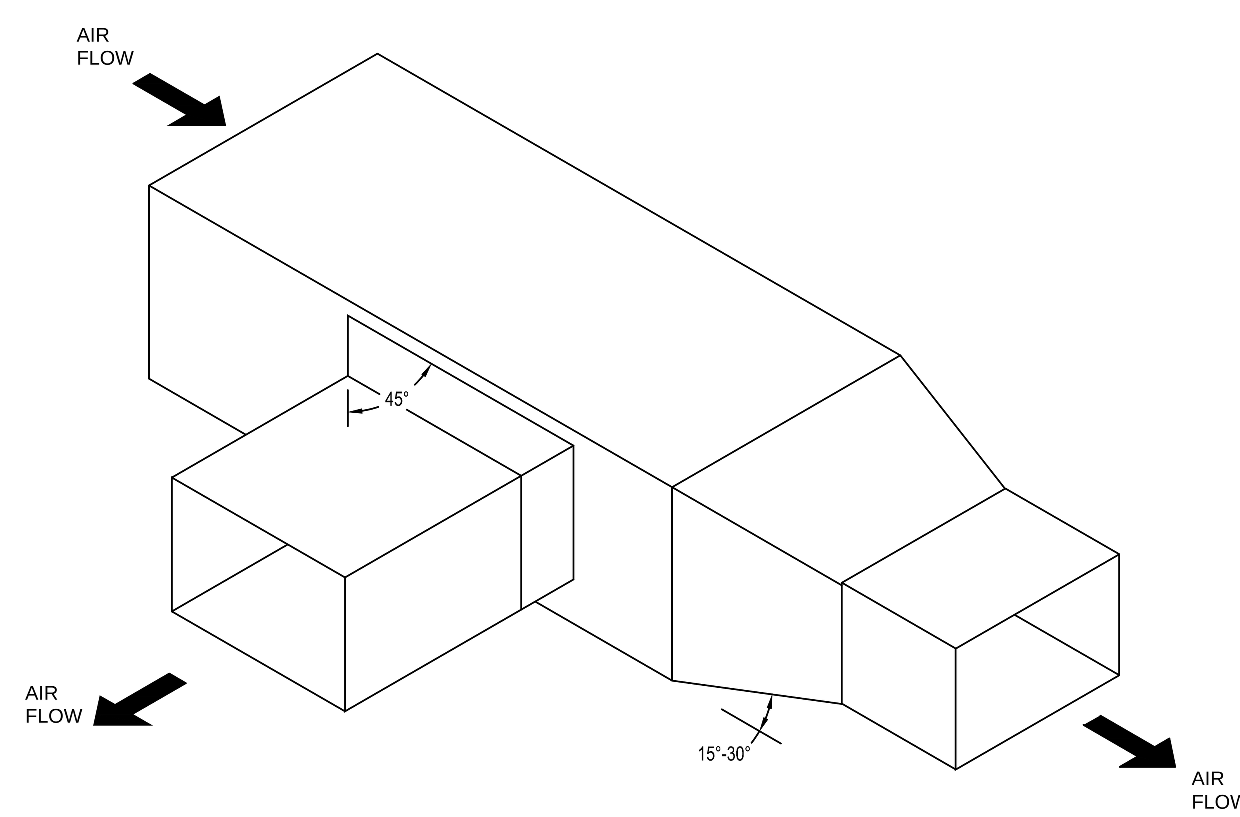
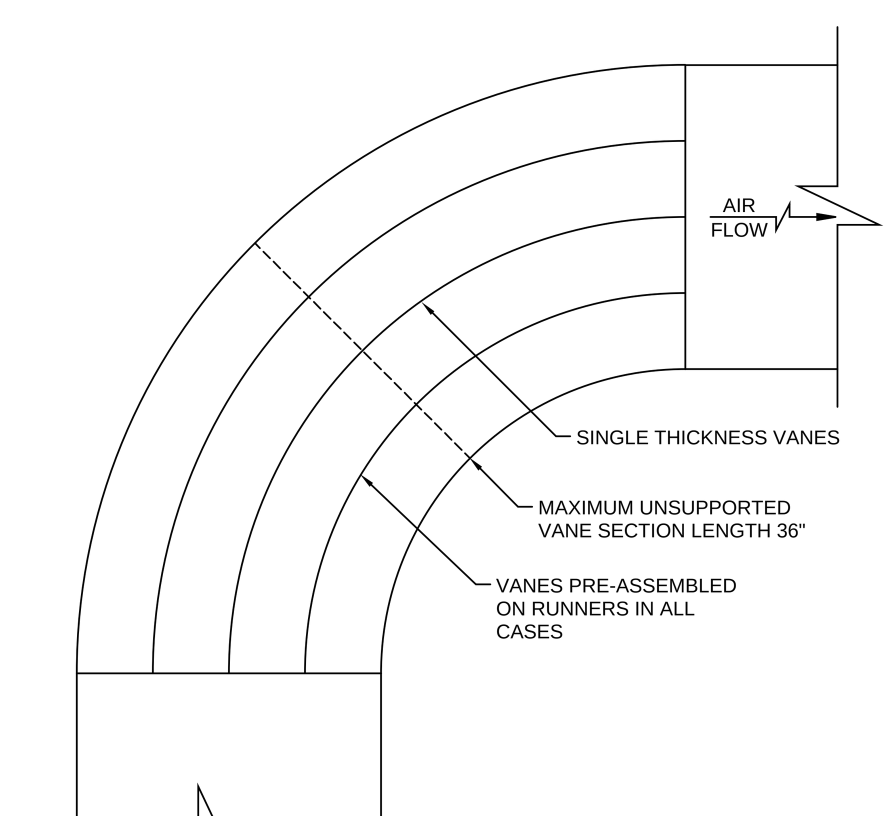


4 DIVERGING TEE, 45° EXIT
SCALE: NO SCALE



1 TYPICAL RADIUS ELBOW WITH VANES
SCALE: NO SCALE

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ISSUE FOR BIDDING	07/02/2018

REVISIONS:

SHEET CONTENTS:
MECHANICAL DETAILS

SHEET NUMBER:
1M601

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SECTION 1 - SUPPLIMENTARY CONDITIONS FOR MECHANICAL WORK

1.1.1 GENERAL CONDITIONS

A. All work covered by this section of these specifications shall be accomplished in accordance with the respective drawings, information of instructions to bidders, general requirements and the supplementary general conditions of these specifications.

B. Bidders shall determine the contents of a complete set of drawings and specifications and be aware that they may be bidding from a partial set of drawings, applicable only to the various separate contract, subcontracts or trades as may be issued for bidding purposes only. The contract documents are the combined Structural, Heating, Ventilating and Air Conditioning and Electrical drawings and specifications. All drawings and specifications are on file in the Engineer's office, and each Bidder shall thoroughly acquaint himself with all of the details of the complete set of drawings and specifications before submitting his bid. All drawings and specifications form a part of the contract documents for each separate contract. They shall be considered as bound therewith in the event partial sets of plans and specifications shall be deemed evidence of the review and examination of all drawings, specifications and addenda issued for this project. No allowances will be made because of the Contractor's unfamiliarity with any portion of the complete set of documents.

C. All equipment and materials shall be manufactured in the United States of America.

1.1.2 SCOPE

A. The work included under this specification consists of the furnishing of all labor, materials, tools, transportation, services, etc. which are applicable and necessary to complete the installation of the systems specified herein; all as described in these specifications, as illustrated on the accompanying drawings, or as directed by the Engineer.

B. In general, the various lines and ducts to be installed by the various trades under this specification shall be run as indicated, as specified herein, as required by particular conditions at the site and as required to conform to the generally accepted standards so as to complete the work in a neat and satisfactorily workable manner. Run work parallel or perpendicular to the lines of the building unless otherwise noted.

C. The construction details for the building are illustrated on the Structural Drawings. Each Contractor shall thoroughly acquaint himself with the details before submitting his bid, as no allowance will be made because of the Contractor's unfamiliarity with these details. Place all inserts to accommodate the ultimate installation of pipe hangers in the forms before concrete is poured. Set sleeves in place in forms before concrete is poured, and in masonry walls while they are under construction. All concealed lines shall be installed as required by the pace of the general construction to precede that general construction.

1.1.3 INSPECTION OF SITE

A. The Contractors shall visit the site, verify all existing items shown on plans or specified herein, and familiarize himself with the working conditions, hazards, existing grades, actual formations, soil conditions, and local requirements involved, and submission of bids shall be deemed evidence of such visit. All proposals shall take the existing conditions into consideration, and the lack of specific information on the drawings shall not relieve the Contractor of any responsibility.

1.1.4 UTILITIES, LOCATIONS AND ELEVATIONS

A. Locations and elevations of the various utilities included within the scope of this work have been obtained from City and/or other substantially reliable sources and are offered separately from the Contract Documents, as a general guide only, without guarantee as to accuracy. The Contractor shall examine the site, shall verify to their own satisfaction the locations, elevations and availability of all utilities and services required and shall adequately inform themselves as to their relation to the work; the submission of bids shall be deemed evidence thereof.

1.1.5 CODE REQUIREMENTS

A. All work shall comply with the provisions of these specifications, as illustrated on the accompanying drawings, or as directed by the Engineer, and shall satisfy all applicable local codes, ordinances, or regulations of the governing bodies, and all authorities having jurisdiction over the work, or services thereto. In all cases where alterations to, or deviations from the drawings and specifications are required by the authority having jurisdiction, the Contractor shall report same in writing to the Owner and secure his approval before proceeding. Upon completion of the work, the Contractor shall provide complete utility service connections, as directed, and submit, as required, all necessary drawings; he shall secure all permits and inspections necessary in connection with his work and pay all legal fees on account thereof. In the absence of other applicable local codes acceptable to the Engineer, the National Electrical Code and International Plumbing Code shall apply to this work.

1.1.6 RECORDS FOR THE OWNER

A. The Contractor shall obtain at his own expense a complete, full-size set of prints on which he shall keep an accurate record of the installation of all materials and systems covered by his contractual agreement. The record shall indicate the location of all equipment and the routing of all systems. All conduit buried in concrete slabs, walls, and below grade shall be located by dimension unless a surface mounted device in each space indicates the exact location. He shall then obtain at his expense one complete reproducible set of the original drawings on which he shall neatly transfer his notations and deliver these drawings to the Engineer at job completion before the final payment for delivery to the Owner.

B. In addition to the above, the Contractor shall accumulate during the job progress the following data in duplicate prepared in a neat brochure or packet folder bonding for subsequent delivery to the Owner. The Contractor shall include in his bid the cost of binding into a book:

1. All warranties, guarantee, and manufacturer's directions on equipment and material covered by the Contract.
2. Copies of approved shop drawings and submittals.
3. Copies of sequence of operations for all equipment covered by Contract.

1.1.7 MATERIALS AND WORKMANSHIP

A. All materials, unless otherwise specified, shall be new, free from any defects and of the best quality of their respective kinds. All like materials used shall be of the same manufacturer, model and quality, unless otherwise specified.

B. All manufactured articles, materials and equipment shall be applied, installed, connected, erected, used, cleaned, adjusted and conditioned as recommended by the Manufacturers, or all indicated in their published literature, unless specifically herein specified to the contrary. All work under this contract shall be performed by competent workmen and executed in a neat and workmanlike manner providing a thorough and complete installation. Work shall be properly protected during construction, including the shielding of soft or fragile materials and the temporary plugging of open lines during construction. At completion, the installation shall be thoroughly cleaned, and all tools, equipment, obstruction or debris present as a result of this Contract shall be removed from the premises.

1.1.8 STORAGE AND PROTECTION

A. Provide adequate facilities for items furnished under these specifications which are subject to damage if exposed to elements. Take such precautions as necessary to properly protect apparatus from damage. Failure to comply with this provision will be sufficient cause for rejection of the particular apparatus involved.

1.1.9 COOPERATION

A. All work under these specifications shall be accomplished in conjunction with other trades on this project in a manner which will allow each trade adequate time at the proper stage of construction to fulfill his work.

B. Maintaining contact and being familiar with the progress of the general construction and the timely installation of sleeves and inserts, etc., before concrete is placed shall be the responsibility of this trade, as will the installation of the required systems in their several stages, at the proper time to expedite this contract and avoid unnecessary delays in the progress of other contracts, and meet all requirements of progress schedules set up by the Engineer.

C. Should any question arise between trades as to the placing of lines, ducts, conduits, fixtures or equipment, or should it appear desirable to remove any general construction which would affect the appearance or strength of the structure, reference shall be made to the Engineer for instruction.

1.1.10 SCHEDULE OF MATERIAL AND EQUIPMENT

A. The Contractor shall submit for approval a complete schedule of material and equipment which is to be installed under the contract. The schedule shall be submitted within 30 days after the award of this contract and prior to the installation or fabrication of any of the material involved. The schedule shall include for materials the Manufacturer's name, Catalog Number, Type and Trade Name; in addition, for equipment, attach Manufacturer's Engineering Data and Specification Sheet.

1.1.11 SHOP DRAWINGS AND SUBMITTALS:

A. Provide Submittals and Shop Drawings (3 copies minimum) for the following equipment and layout:

1. Ductwork fabrication details and layout at 1/8" = 1'-0" scale.
2. Mechanical equipment cut sheets including all performance characteristics, accessories, drawings, wiring diagrams, etc. Accessories shall be clearly labeled to show what is and is not provided.
3. Piping details showing materials used and joining/sealing methods.
4. Piping layout at 1/8" = 1'-0" scale.

B. Equipment shall not be ordered until approved by the Engineer of Record. The Contractor shall allow two (2) weeks for design team review of submittals.

1.1.12 DRAWINGS AND SPECIFICATIONS

A. The drawings show, diagrammatically, the locations of the various lines, ducts, conduits, fixtures and equipment and the method of connecting and controlling them. It is not intended to show every connection in detail and all fittings required for a complete system. The systems shall include, but are not limited to, the items shown on the drawings. Exact locations of these items shall be determined by reference to the general plans and measurements at the building and in cooperation with other sub-contractors and, in all cases, shall be subject to the approval of the Contractor. the Contractor reserves the right to make any reasonable change in the location of any part of this work without additional cost to the Owner.

B. Should any changes be deemed necessary by the Contractor in items shown on the contract drawings, shop drawings and descriptions, the reason for the proposed changes shall be submitted to the Owner for approval.

C. Exceptions and inconsistencies in plans and specifications shall be brought to the contractor's attention before bids are submitted; otherwise, the Contractor shall be responsible for the cost of any and all changes and additions that may be necessary to accommodate his particular apparatus.

D. The Contractor shall lay out his work maintaining all lines, grades and dimensions according to these drawings with due consideration for other trades and verify all dimensions at the site prior to any fabrication or installation. Should the layout be impractical, the Contractor shall be notified before any installation or fabrication, and the existing conditions shall be investigated and proper changes effected without any additional cost.

E. Titles of Sections and Paragraphs in these specifications are introduced merely for convenience and are not to be construed as a correct or complete segregation to tabulation of the various units of material and/or work. The Engineer does not assume any responsibility, either direct or implied, for omissions or duplications by the Contractor or any Sub-contractor due to real or alleged error in the arrangement of matter in the Contract Documents.

1.1.13 ENGINEER'S APPROVAL

A. In any statement under this contract where "approval" is required or requested, it is understood that such approval must be obtained from the Engineer in writing before proceeding with the proposal, and an adequate number of copies of any such proposal shall be submitted to the Engineer.

B. The approval by the Engineer of any materials, changes, drawings, etc., submitted by the Contractor will be considered as general only and to aid the contractor in expediting his work. such approval as may be given does not in any way relieve the Contractor from the necessity of furnishing the materials and performing all work as required by the drawings and specifications.

1.1.14 LOCAL RESTRICTIONS

A. The Contractor shall become familiar with all rules and regulations of the City, County and State, or any other authority having jurisdiction over this project. If it is the Contractor's opinion that any work or materials shown on the drawings or specifications do not comply with these rules and regulations as to size, type, capacity and quality, he must make it known prior to the submission of his bid, which shall be deemed evidence of compliance; otherwise, the Contractor shall be responsible for the approval of all work or material and, in the event that such Authority should indicate disapproval, he shall correct same with materials approved by the Engineer at no additional cost to the Owner.

1.1.15 ELECTRICAL WIRING

A. Except for such items as are normally wired up at their point of manufacture and so delivered, and unless specifically noted to the contrary herein, the Electrical Subcontractor will do all electric wiring of every character for power supply. The Mechanical Subcontractor shall erect all motors in place ready for connections and shall furnish with each such motor a starter of the type specified and deliver it in good condition to the Electrical Subcontractor at the job. The Electrical Subcontractor will mount all such starters, as directed, furnishing supporting structures where necessary. The Owner and other Subcontractors shall furnish with each item requiring electrical connections, the necessary instructions and wiring diagrams to the Electrical Subcontractor. The Electrical Subcontractor shall refer to the Specifications to determine the Scope of the Work.

1.1.16 LARGE APPARATUS AND EQUIPMENT

A. All large apparatus and equipment which is specified or shown to be furnished or installed under this Contract, and which may be too large to be moved into its final position through the normal building openings planned, shall be placed by this Subcontractor in its approximate final position. This shall be accomplished through cooperation and coordination with other Subcontractors before any obstructing structure is installed. All apparatus shall be cribbed up from the floor by this Subcontractor and cared for as specified under "Storage and Protection" or as directed by the Engineer.

1.1.17 RESPONSIBILITY

A. The Contractor will be held responsible for the satisfactory and complete execution of all work included. He shall produce complete finished operating systems and provide all incidental items required as part of his work, regardless of whether such item is particularly specified or indicated.

1.1.18 CLEAN UP

A. Clean up trash and debris caused by the work of this Section, keeping premises, streets, sidewalks and adjacent areas clean and neat at all times. B. Dispose of such materials outside the limits of the project site to approved locations.

1.1.19 PAINTING

A. Upon completion, clean all pipes before painting. Painting of piping shall be in accordance with the Owner's color standards

1.1.20 ACCESS DOORS

A. Access doors are to be provided by the Contractor. Contractor will closely coordinate locations of valves, etc. in order to have access to all concealed portions of the system requiring periodic service. Prepare shop drawings for coordination of all access doors, locating same for installation by General Contractor. Access door locations shall be approved by Engineer or Owner before installation.

1.1.21 FLAME SPREAD PROPERTIES OF MATERIALS

A. All materials and adhesives used for acoustical linings and insulation, jackets, tapes, etc. shall conform to Interim Federal Standard Flame-spread Properties of Materials, Inc. Fed. Std. No. 00336A (comm. NBS). The classification shall not exceed No. 2, with the range of indices between 0 and 25 for these classifications as listed in the Federal Specifications for the basic materials, the finishes, adhesives, etc. specified for each system, and shall be such that when completely assembled the total will not exceed an index of 50 in Classification 111 as listed in the Federal Specifications. Modifications shall be made to insulating materials, etc. as required to comply with the Federal Specifications.

1.1.22 GUARANTEE

A. The Contractor shall furnish a written guarantee in triplicate, warranting all materials, equipment and labor furnished by him to be free of all defects for a period of one year from date of final acceptance by the Owner. He shall further guarantee that all equipment shall meet the characteristics, capacities and workmanship specified and within the warranty period, the defects and/or equipment will be repaired or made good without cost to the Owner. The Contractor further agrees to correct warranty deficiencies within 48 hours of notification by management.

B. REFERENCE DOCUMENTS: Conditions of the Contract and Division 01 "General Requirements" are made a part of this section whether attached hereto or not.

SECTION 4 - HEATING, VENTILATION AND AIR-CONDITIONING SYSTEMS

4.1.1 SCOPE

A. Provide complete air supply, return, outside air and exhaust systems including fans, terminal devices and other components specified herein.

4.1.2 SUBMITTALS

A. Shop Drawings: Submit complete shop drawings, in accordance with Section 1, indicating materials, quantities, sizes and installation details.

4.1.3 COORDINATION

A. Install materials and equipment at proper time to keep pace with the general construction and the work of the other trades involved.

4.1.4 WARRANTY

A. The Mechanical Sub-contractor shall warranty all material, workmanship and equipment for a period of one year after final acceptance by the Owner. The warranty specifically implies that any defective portion becoming apparent during this period will be repaired, replaced or otherwise made good at no additional cost to the Owner. It shall further include replacement or refrigerant loss not due to Owner negligence. Compressors shall contain an additional four-year warranty.

4.2.1 DUCTWORK

A. Rigid Ductwork: All air conditioning and exhaust ductwork, plenum, casings and sheet metal, connections shall be fabricated of new joint-forming quality galvanized prime grade sheets.

B. Rectangular Low Pressure Ducts: Constructed of the following minimum gauges:

Largest Dimension of Duct	Gauge of Metal
Up to 12"	No. 26 U.S. Gauge
13" to 30"	No. 24 U.S. Gauge
31" to 54"	No. 22 U.S. Gauge

C. Round Low Pressure Ducts: "SNAP-LOK" as manufactured by United Sheet Metal Company.

D. Rectangular Ductwork Fittings: Fabricated per SMACNA Standards for low-pressure ductwork(2-inch pressure class).

E. Round Ductwork Fittings: As manufactured by United Sheet Metal Co., and/or as detailed on the drawings.

F. Flexible Connections: Connections to air conditioning units and fans shall be flexible connections which shall be neoprene coated glass fabric weighing not less than 30 ounces per square yard and at least 1/16" thick.

G. At the Contractor's option, 2" insulated flexible duct may be used for final run out to air devises when installed per manufacturer's installation instructions. Flexible run outs shall not exceed 5-feet extended length.

H. Surface-burning characteristics for sealants and gaskets shall be a maximum flame-spread index of 25 and a maximum smoke-developed index of 50 when tested according to UL 723; certified by an NRTL

I. Access doors shall be provided for access to all dampers, fusible links, and where required for maintenance and cleaning operations. Access doors serving insulated ducts shall be double-skin doors with one inch of insulation on the door. Where duct size permits, the access doors shall be16-inches by 18-inches. Access doors shall be as manufactured by MILCOR.

4.3.1 INSULATION

A. All rectangular sheet metal ducts shall be insulated with 1.5-inch" thick, 3/4" lb density fiberglass-faced insulation, or as required to meet a minimum installed R-value of 5.0. Install with all joints overlapped and neatly sealed.

B. All round sheet metal ducts shall be insulated with 2" thick, 3/4" lb density fiberglass-faced insulation, or as required to meet a minimum installed R-value of 5.0. Instal with all joints overlapped and neatly sealed with UL 181 listed sealant.

B. Insulate refrigerant piping with 3/8" thick ARMAFLEX. Apply insulation with all joints firmly butted together.

SECTION 5 - SYSTEM BALANCING

5.1.1 SCOPE

A. Testing, adjustment and start-up of mechanical systems shall be performed by Air Balancing Company (Contact: Bret Privitt, bprivitt@airbalancingco.com, (817)572-6994). All necessary test equipment, instruments, materials and labor required for performing all the tests described shall be provided as part of the work of this division.

B. Upon completion of the installation and start-up of the mechanical equipment, check, adjust and balance systemic components to obtain optimum conditions in each conditioned space in the building.

C. Prior to requesting a final inspection, this sub-contractor shall prepare and submit to the engineer of record complete reports on the balance and operations of the system, bearing the seal of a certified air balance technician. In this report, the original conditions measured at startup and final conditions after balancing of all equipment shall be clearly indicated.

D. Make an inspection in the building during the opposite season from that in which the initial adjustments were made and, at the time, make any necessary modifications to the initial adjustments required to produce optimum operation of the systemic components to produce the property conditions in each conditioned space.

5.1.2 WORK INCLUDED

A. The balancing technician shall be responsible for inspecting, adjusting, balancing and logging the data on the performance of fans, all dampers in the duct systems and all air distribution devices. The mechanical contractor and the suppliers of the equipment installed shall all cooperate with the balancing technician to provide all necessary data on the design and proper application of the systematic components and shall furnish all labor and materials required to eliminate any deficiencies or improper-performance.

B. During the balancing, the temperature regulation shall be adjusted for proper relationship between controlling instruments and calibrated by the temperature controls sub-contractor using data submitted by the balancing technician. The total variation shall not exceed 3 degrees from the present median temperature during the entire temperature survey period.

C. In all fan systems, balance the air quantities to be between plus 10- to minus 5-percent of the values shown on the plans. It shall be the obligation of the mechanical contractor to furnish or revise fan drives and/or motors, if necessary, without cost to the contractor, to attain the specified air volume.

5.1.3 REPORT

A. Before final acceptance is made, the balancing technician shall prepare a detailed, written report.

B. The data shall be neatly entered on appropriate forms together with any typed supplements required to completely document all results.

C. Written explanations of any abnormal conditions shall be included. All this shall be assembled into a suitable brochure, and a total of four copies shall be provided.

D. The typed test data sheets and correlation of the test results shall be certified to be true and correct by a certified air balance technician over the signature of the subcontractor. Such signature shall be executed by an officer if the subcontracting firm is a corporation, a partner if a partnership, or by the owner is a sole ownership. This data shall be delivered to designated members of the building operating personnel not less than three days after the texts are complete settings, reading, etc. shall be prepared and submitted in quadruplicate.

5.1.4 INSTRUCTIONS

A. During the test periods, the balancing technician shall instruct the building maintenance personnel in the construction and operation of all equipment.

SECTION 6 - PIPING FOR MECHANICAL SYSTEMS

A. Piping Materials:
  1. Schedule 40, standard weight, seamless black steel - ASTM A53 or ASTM A106
  2. Use for:
  a. Chilled water 2.5-inch up to 10-inch
  b. Heating water 2.5-inch up to 10-inch
  3. Type M, hard drawn, seamless, wrought copper - ASTM B88
  a. Use for: Cold condensate drain

B. Fittings and Joints:
  1. Steel:
  a. All sizes and service: Mark in accordance with MSS-SP-25
  b. Unions up to 2.5-inch: Same material type as fitting; screwed with brass seat
  c. Branch connections from mains and headers, 2.5-inch and larger: Welded tees or welded outlets (Bonney Forge Weldolets or Threadolets). Use forged outlets only if branch line is at least one pipe size smaller than main or header.
  2. Galvanized steel: Same as for steel except galvanized coating
  3. Copper:
  a. Sizes and service noted in piping materials above:
  b. Sweat type, wrought copper, ASTM B62 with dimensions conforming to ANSI B16.22 and sweep patterns for copper tubing.
  c. Provide dielectric connections at junctions of copper pipe/equipment connections and steel pipe systems.
  d. Provide copper solder joint to plated female iron pipe.
  e. Unions - Brass ground joint, 250-lb working pressure
  f. Nipples - Brass

C. Piping Insulation:
  4. Provide factory pre-molded, segment type insulation for pipe, valves and fittings. Fitting insulation shall be the same thickness and material as adjoining material.
  5. Surfaces shall be clean and dry when covering is applied. Covering to be dry when installed, before and during application of any finish - unless such finish specifically requires a wetted surface for application.
  6. Adhesives, cements and mastics shall be compatible with materials applied and shall not attack material in either wet or dry state.
  7. For pipe fittings, valves, strainers and other irregular surfaces, in chilled water for refrigerant systems operating below 60-degrees f, when inside building or in equipment rooms, cover insulation with white colored woven glass fabric embedded in white vapor barrier coating, Foster 30-35 or equal.
  8. Fiberglass - Johns-Manville "MICRO-LOK 850" or equal, ASJ fiberglass reinforced kraft paper with aluminum foil, minimum installed R-value of 5.5:
  a. Use for:
    - Chilled water - All sizes (1.5-inches)
    - Heating water - 1.5-inch and less (1-inch)
    - Heating water - 2-inch and larger (1.5-inches)
    - Cold condensate; above ground, indoors - All sizes - (1-inch)

D. Valves:
  1. Valve connections - 2-inch and smaller - threaded; 2.5-inch and larger - flanged
  2. All balancing valves and butterfly valves shall be provided with position indicators and maximum adjustable stops (memory stops).
  3. All valves (except control valves) and strainers shall be full size of pipe before reducing size to make connections to equipment and controls.
  4. Unions and/or flanges shall be installed at each piece of equipment, in bypasses and in long piping runs (100-feet or more) to permit disassembly for alterations and repairs.
  5. Check
    a. HVAC water; 125 psi operating pressure and less - NIBCO Figure 910, non-slam
  6. Ball
    a. HVAC water; 2-inch and less - NIBCO Figure T-585-70 or S-585-70, 2-piece, full port, 600 psi, WOG, TFE seats
    b. Ball valves shall have locking handles. provide extended lever for insulated service.
  7. Butterfly
    a. HVAC Water; 2.5-inch and up - NIBCO Figure LD-2000, lug type, 200 PSI, Class 125, EPDM liner, aluminum bronze disc
    b. Butterfly valves shall be rated bubble-tight for dead-end service at full pressure in both directions without the need for downstream blind flange.
  8. Plug
    a. 2-inch and less - NORDSTROM Figure #114 lubricated plug cock
    b. 2.5-inch to 4-inch - NORDSTROM Figure #115 lubricated plug cock
    c. 5-inch and up - NORDSTROM Figure #169 lubricated plug cock
    d. Provide visual position indicator.

E. Hangers:
  1. Non-insulated steel and galvanized piping 2-inch through 24-inch - GRINNELL Figure #260 MSS Type 1 clevis hangers
  2. Steam and steam condensate return 3-inch and less; heating water 4-inch and less; all other insulated piping - GRINNELL Figure #260 clevis hangers with Figure #167 MSS Type 40 galvanized insulation protection shields sized for supporting insulation having a compressive strength of 4 psi.
    a. Support piping on outside of insulation.
    b. Size hangers so that pipe insulation passes through them without interruption.
  3. Non-insulated copper tubing with no longitudinal movement - GRINNELL Figure #ct-69 MSS Type 10 with adjustable copper-plated wrought tubing ring hanger.
  4. Use HILTI KWIK-BOLT, zinc plated, metal expansion anchors. anchor to meet UL, ICBO-4627 and FM listings.
  5. Attachments to bar joists and steel beams - GRINNELL Figure #218, MSS Type 30 malleable beam clamps
  6. Minimum hanger spacing:
    a. Steel pipe - Water
    - 0.25-inch to 1.5-inch - 7'-0"
    - 2-inch to 2.5-inch - 10'-0"
    - 3-inch and larger - 12'-0"
    b. Copper pipe - Water
    - 0.25-inch to 1.5-inch - 5'-0"
    - 2-inch to 2.5-inch - 8'-0"
    - 3-inch and larger - 10'-0"
  7. Provide minimum of one hanger per section close to joint at change of direction and at branch connections.

F. Installation:
  1. Piping connections to water coils shall be fabricated with isolation valves, flanges and/or unions positioned to allow removal and service of the component parts.
  2. Thermometer wells shall be installed on top or side of horizontal piping in order to retain gauge fluid and to be easily read from the floor.
  3. Pressure gauges shall be installed on top or side of horizontal piping in order to be easily read from the floor.
  4. Install piping neatly, orderly and parallel to building lines.
  5. Unless otherwise noted, chilled and heating water run outs to terminal units shall be minimum 0.75-inch diameter.
  6. Provide an air vent at the high point of each drop in the heating water, chilled water and other closed-loop water piping systems.
  7. Unless otherwise noted, all piping is overhead, tight to underside of structure, or slab with space for insulation, if required.
  8. Install piping so that all valves, strainers, unions, traps, flanges and other appurtenances requiring access are accessible.
  9. All valves shall be installed so that valve remains in service when equipment or piping on equipment-side of valve is removed.
  10. Install all piping without forcing or springing.
  11. All piping shall clear doors and windows.
  12. All valves shall be adjusted for smooth and easy operation.

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STATE OF TEXAS
7/2/2018
RENÉ A. CULROSS
71739
LICENSED
PROFESSIONAL ENGINEER
9438.7.10.2018.16184

Tarrant County Plaza Building
Air Handler Replacement
Phase 1

Tarrant County Plaza
200 Taylor St, Fort Worth, TX 76102

TARRANT COUNTY TEXAS

JOB NUMBER: P16184

DATE ISSUED: JULY 02, 2018

ISSUES:

ISSUE FOR REVIEW	03/24/2017
ISSUE FOR PRICING	03/31/2017
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REVISIONS:

SHEET CONTENTS:
MECHANICAL SPECIFICATIONS

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